

QUANTUM AUTOMATION PROJECT
NOTES AND AREAS FOR FUTURE EXPLORATION.

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I- REVIEW MATERIAL:

A- SMOOTH MANIFOLDS AND FORMS

For purposes of brevity, we'll only be writing summaries of what is necessary to write this paper.

A smooth n -manifold is a Hausdorff and, for our purposes, second countable topological space M that $\forall p \in M \exists U \in \mathcal{N}(p); U \cong \mathbb{R}^n$

Let M be a topological n -manifold. A coordinate chart on M is a pair (U, φ) , where U is an open and $\varphi: U \rightarrow \hat{U} \subset \mathbb{R}^n$ is a homeomorphism from U to an open subset $\hat{U}$ of $\mathbb{R}^n$.

An atlas is a collection of charts covering M , the atlas is smooth whenever two charts overlap,

$$(U, \varphi) \cap (V, \psi)$$

The transition map

$$\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$$

Is smooth.

Let M be a smooth manifold A function $f: M \rightarrow \mathbb{R}$ Is smooth if, in every chart

$$f \circ \varphi^{-1}: \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

Is smooth in the usual multivariable calculus sense.

More generally, if $F: M \rightarrow N$ is a map between manifolds and F is smooth if

$$\psi \circ F \circ \varphi^{-1}$$

Is a smooth map between open subsets of Euclidean spaces.

For tangent vectors

A tangent vector is an operator that differentiates functions

A **vector field** X on a manifold M is a smooth assignment

$$p \mapsto X_p \in T_p M$$

— at each point you get a tangent vector. So evaluating X "at p " gives you a single tangent vector living in the tangent space $T_p M$.

Equivalently (and this is the more useful definition for calculations), a vector field is a **derivation** on smooth functions: X eats a function $f: M \rightarrow \mathbb{R}$ and gives a new function Xf , where

$$\left(X(f)(p) = X_p(f) = \frac{d}{dt} \Big|_{t=0} f(\gamma(t)) \right)$$

for any curve γ with $\gamma(0) = p$, $\dot{\gamma}(0) = X_p$. In coordinates,

$$X = \sum_i X^i(x) \frac{\partial}{\partial x_i}.$$

The basis vectors $\partial / \partial x_i$ at a point p form a basis of $T_p M$.

More precisely

Suppose that (U, φ) is a chart, $\forall p \in U$

$$\varphi(p) = (x^i(p))_{i \in \{1, \dots, n\}}$$

Then a tangent vector at p , $v \in T_p M$, would be something of the form

$$v = (v^1, \dots, v^n) \cdot \left(\frac{\partial}{\partial x^i} \right) \Big|_p = \sum_{i=1}^n v_i \frac{\partial}{\partial x^i} \Big|_p$$

For example, if we have a regular surface in $\mathbb{R}^3$, parametrized by

$$\sigma(u, v) = (x^1(u, v), x^2(u, v), x^3(u, v))$$

Then a vector in the tangent space $X \in T_p \Sigma$, $p = \sigma(u_0, v_0)$ would look like

$$X = v_1 \sigma_u(u_0, v_0) + v_2 \sigma_v(u_0, v_0)$$

More precisely $T_p \Sigma = \text{span}\{\sigma_u, \sigma_v\}|_p$

The Cotangent space at p is the dual vector space of $T_p M = \text{Hom}(T_p M, \mathbb{R})$

They're what gives us forms, differential forms that is precisely:

recall that the basis of $T_p M = \left(\frac{\partial}{\partial x^i}\right)_{1 \leq i \leq n}$ then the dual basis of T_p^* would be

$$T_p^* M = \langle dx^i \rangle_{1 \leq i \leq n}$$

That satisfy $dx^i \left(\frac{\partial}{\partial x^j}\right) = \delta_j^i$

$$v = \sum_{j=1}^n \frac{v^j \partial}{\partial x^j} \Rightarrow dx^i(v) = v^i$$

Recall from multivariate calculus that the differential of a 0-form creates a 1-form, and basically this is what df entails if $f: M \rightarrow \mathbb{R}$ is smooth, then the differential of f at p is the covector

$$df_p: T_p M \rightarrow \mathbb{R}$$

$$df_p(v) = v(f)$$

So df_p eats a tangent vector and returns the directional derivative of f in that direction

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$

In Euclidean space, people identify df with ∇f , but this secretly uses the "*Euclidean metric*"

a **1-form** α is, pointwise, an element of the **cotangent space** $T_p^* M$, which is by definition the dual of the tangent space:

$$T_p^* M = (T_p M)^* = \{\text{linear maps } T_p M \rightarrow \mathbb{R}\}.$$

So at each point, a 1-form is a linear functional on tangent vectors. Globally, a 1-form is a smooth assignment $p \mapsto \alpha_p \in T_p^* M$, and it pairs with a vector field to produce a function:

$$\alpha(X): p \mapsto \alpha_p(X_p) \in \mathbb{R}.$$

The dual basis to $\partial/\partial x_i$ is written dx^i , characterised by

$$dx^i \left(\frac{\partial}{\partial x_j} \right) = \delta_j^i.$$

So in coordinates a 1-form looks like $\alpha = \sum_i \alpha_i(x) dx^i$, and it eats a vector field $X = \sum_j X^j \partial_j$ by

$$\alpha(X) = \sum_i \alpha_i X^i.$$

That's literally a row vector dotted with a column vector.

Where does **df** come from?

If $f \in C^\infty(M)$, the **differential** df is the 1-form defined by

$$df(X) = X(f).$$

In coordinates $df = \sum_i (\partial f / \partial x_i) dx^i$. So 1-forms generalise gradients, except they don't need a metric to exist — they're intrinsically dual objects.

A **k -form** ω eats k tangent vectors at each point and produces a number, **antisymmetrically and multilinearly**:

$$\omega_p : \underbrace{T_p M \times \cdots \times T_p M}_{k \text{ copies}} \rightarrow \mathbb{R}.$$

Pointwise, the space of such gadgets is $\Lambda^k(T_p^* M)$ — the k -th exterior power of the cotangent space. Built from wedges of 1-forms:

$$(dx^i \wedge dx^j)(u, v) = u^i v^j - u^j v^i,$$

which is exactly the signed area of the projected parallelogram. A symplectic form is then a 2-form, and for example, $\omega_0 = \sum dx_i \wedge dy_i$ lives in $\Lambda^2(T^* \mathbb{R}^{2m})$ at every point.

Object	Pointwise lives in	Pairs with
Vector field X	$T_p M$	functions (gives Xf)
1-form α	$T_p^* M = (T_p M)^*$	vector fields (gives a number)
k -form ω	$\Lambda^k T_p^* M$	k vector fields (gives a number)
metric g	$T_p^* M \otimes T_p^* M$ (symmetric)	2 vector fields $\rightarrow$ number
symplectic ω	$\Lambda^2 T_p^* M$	2 vector fields $\rightarrow$ signed area

The "duality" is **pointwise**. There's no canonical isomorphism $T_p M \cong T_p^* M$ without extra structure. A **metric** g gives you one (the musical isomorphisms $\flat: X \mapsto g(X, \cdot)$ and its inverse $\sharp$) — that's how you turn df into ∇f . A **symplectic form** also gives you one, via $X \mapsto \omega(X, \cdot)$ — and *that* is what defines the Hamiltonian vector field X_H from dH .

This is the conceptual core: a metric and a symplectic form are both ways of identifying vectors with covectors, just with different symmetry properties — and on a Kähler manifold, they're both there at once and compatible.

A nice example would be the following

Let

$$H_N = \text{span}\{e_1, \dots, e_N\} \subset L^2(0, \pi)$$

Where

$$e_k(x) = \sqrt{\frac{2}{\pi}} \sin(kx)$$

Every element $u \in H_N$ has the form

$$u = a_1 e_1 + \cdots + a_N e_N$$

We define

$$M_N = \{u \in H_N : \|u\|_{L^2} = 1\}$$

Define

$$\Phi: H_N \rightarrow \mathbb{R}, \quad \Phi(u) = \|u\|_{L^2}^2$$

Then

$$M_N = \Phi^{-1}(1).$$

The derivative of Φ at u in the direction of $h \in H_N$ is

$$D\Phi(u)(h) = 2(u, v)_{L^2}$$

If $u \in M_N$, then $u \neq 0 \Rightarrow D\Phi_u \neq 0$, then 1 is a regular value of Φ , and M_N is a smooth $(N - 1) - \text{manifold}$.

$$T_u M_N = \{v \in H_N : v \in \ker D\phi_u\} = \{v \in H_N : \langle u, v \rangle_{L^2} = 0\}$$

Now consider the functional

$$F: M_n \rightarrow \mathbb{R}$$

$$F(u) = \int_0^\pi q(x) u(x)^2 dx$$

Then

$$dF_u(v) = 2 \int_0^\pi q(x) u(x) v(x) dx$$

Observe, we are working in

$$H_N = \text{span}\{e_1, \dots, e_N\}$$

So, every $u \in H_N$ is

$$u = \sum_{k=1}^N a_k e_k$$

$$u \rightarrow (a^1(u), \dots, a^N(u))$$

So that our coordinate chart is $a^i(u) = \langle u, e_i \rangle_{L^2}$

And hence our $T_u^*M_N = \text{span}\{da_1, \dots, da_n\}$

$$\begin{aligned} 2 \int_0^\pi q(x) u(x) v(x) dx &= 2 \int_0^\pi q(x) \sum_i a_i e_i(x) \sum_j v^j e_j(x) dx \\ &= 2 \sum_{i,j} a_i v^j \left[\int_0^\pi q(x) e_i e_j dx \right] \\ &= 2 \sum_{i,j} a_i Q_{ij} v^j = a^t [Q] v \end{aligned}$$

Notice how $\frac{\partial F}{\partial a_j} = 2 \sum_i a_i Q_{ij}$, then

$$dF = a^t [Q] = \sum_{j=1}^N 2 \frac{\partial F}{\partial a_j} da_j$$

At a point $p \in M$, a k -form is an alternating multilinear map

$$\omega_p: \prod_{1 \leq i \leq n} T_p M \rightarrow \mathbb{R}$$

It takes k tangent vectors as input and returns a number. Geometrically:

A 1-form measures oriented length along a direction.

A 2-form measures oriented area... a 3-form measures oriented volume...

In local coordinates

$$(x^1, \dots, x^n),$$

a 1-form looks like

$$\alpha = \sum_{i=1}^n a_i dx^i$$

A n -form looks

$$\omega = \sum_{i_1 < \dots < i_k} w_{i_1 \dots i_k} \bigwedge_{i=1}^k dx^{i_i}$$

The space of smooth k-forms on M is denoted

$$\Omega^k(M).$$

The wedge product combines forms, if $\alpha \in \Omega^k(M), \beta \in \Omega^\ell(M)$ then

$$\alpha \wedge \beta \in \Omega^{k+\ell}(M)$$

If $\dim M = n$, then to integrate over all of M, you need something that measures - dimensional volume, which is precisely an n-form.

On a Riemannian manifold, the metric gives a natural volume form

$$dvol_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n$$

So we can integrate

$$\int_M f dvol_g$$

Suppose $F: N \rightarrow M$ is a smooth map, if $\omega \in \Omega^k(M)$, it's pullback is a k-form

$$F^* \omega \in \Omega^k(N)$$

At a point $p \in N$, we define

$$(F^* \omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(dF_p(v_1), \dots, dF_p(v_k))$$

Pullback lets us integrate forms using parametrizations if

$$\Phi: U \subset \mathbb{R}^k \rightarrow M$$

Parametrizes a k-dimensional piece of M, then

$$\int_{\Phi(U)} \omega = \int_U \Phi^* \omega$$

So pullback is the rigorous version of substitution

Examples:

Take the Finite Fourier space

$$H_N = \text{span}_{\mathbb{C}}\{e_{-N}, \dots, e_N\} \subset L^2(S^1; \mathbb{C}),$$

Where

$$e_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}.$$

So, every function is

$$u(x) = \sum_{k=-N}^N z_k e_k(x), \quad z_k \in \mathbb{C}$$

Hence $H_N \cong \mathbb{C}^{2N+1}$, Now impose the L^2 – *unit* constraint:

$$\mathcal{S}_N = \{u \in H_N : \|u\|_{L^2} = 1\}$$

we'll show that it's a **real** manifold, it important to distinguish it from it being a complex manifold, which is a totally different thing, (the defining equation must be independent of $\bar{z}$ *for holomorphicity*)

because the Fourier basis is orthonormal

$$\|u\|_{L^2}^2 = \sum_{n=-N}^N |z_k|^2$$

So defining

$$\Phi: \mathbb{C}^m \rightarrow \mathbb{R}, \quad \Phi(z) = \sum_{k=-N}^N |z_k|^2$$

Write $z_k = a_k + ib_k$, $\Phi(z) = \sum_{k=-N}^N a_k^2 + b_k^2$

$$d\Phi_z(\zeta) = 2\operatorname{Re}\langle z, \zeta \rangle_{\mathbb{C}^m}$$

For $z \in S_N$, $d\Phi_z \neq 0_{\operatorname{Hom}(H_N, \mathbb{R})}$, making it a submersion and giving us a smooth real submanifold of $\mathbb{C}^m$ of codimension 1.

Thus $\mathcal{S}_N$ is a smooth manifold of real dimension, with $T_z S_N = \{\zeta \in \mathbb{C}^m : \operatorname{Re}\langle z, \zeta \rangle = 0\}$

Writing the 1-forms, each complex coordinate is

$$z_k = a_k + ib_k$$

So the real coordinate 1-forms are da_k & db_k

With

$$dz_k = da_k + idb_k$$

$$d\bar{z}_k = da_k - idb_k$$

A basic Kahler-style 2-form on $\mathbb{C}^m$ is

$$\omega_0 = \frac{i}{2} \sum_k dz_k \wedge d\bar{z}_k$$

But $dz_k \wedge d\bar{z}_k = (da_k + idb_k) \wedge (da_k - idb_k) = -2ida_k \wedge db_k$

$$\omega_0 = \sum_k da_k \wedge db_k$$

Now consider the inclusion:

$$\iota: S_N \hookrightarrow \mathbb{C}^m$$

What the pullback $\iota^*\omega_0$,

For a smooth map $f: M \rightarrow N$, between manifolds and a k -form ω , the pullback $f^*\omega$ is the k -form on M defined pointwise by

$$(f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(df_p(v_i))$$

Computationally

$$(\iota^*\omega_0)_p(u, v) = \omega_0|_p(u, v) = u, v \in T_p S_N$$

It remains the same, but we only evaluate it on vectors that are tangent to the sphere, what changes is the domain on which we're allowed to evaluate it.

This is the general principle of pulling back along an embedding: a pulled-back form is the original form restricted to the tangent directions of the submanifold.

On all of $\mathbb{C}^m$, ω_0 is nondegenerate, however pulling back to the sphere, this is no longer the case, there is a tangent direction along which ω_0 vanishes against everything tangent. That direction is exactly what the image is identifying.

The phase action. The circle group $S^1 = \{e^{i\theta}\}$ acts on $\mathbb{C}^m$ by

$$\theta \cdot z = e^{i\theta} z$$

This action preserves the norm, so the vector remains on the sphere.

The vector field generating the phase rotation is $z \rightarrow iz$, so $iz \in T_z \mathcal{S}_N$, this direction has a name: the phase direction, or the Hopf direction.

$$(\iota^* \omega_0)_z(iz, v) = \text{Im}\langle iz, v \rangle = -\text{Re}\langle z, v \rangle = 0$$

So iz is in the kernel of $\iota^* \omega_0$ at every point. The pulled-back form is **degenerate**, with a 1-dimensional kernel given exactly by the phase direction $\mathbb{R} \cdot iz$, it's a general fact that pulling a symplectic form back to an odd-dimensional submanifold must produce a degenerate form (symplectic forms only exist in even dimensions), and the kernel is called the *characteristic foliation*.

The quotient: complex projective space

Because the phase direction is the kernel of $\iota^* \omega_0$, if you collapse it (i.e., quotient the sphere by the phase S^1 -action), the form descends to a *nondegenerate* 2-form on the quotient. The quotient is

$$\mathcal{S}_N/S^1 = \mathbb{CP}^{m-1},$$

complex projective space of complex dimension $m - 1$. This is the **Fubini–Study symplectic manifold**: there is a unique 2-form ω_{FS} on $\mathbb{CP}^{m-1}$ such that $\pi^* \omega_{FS} = \iota^* \omega_0$, where $\pi: \mathcal{S}_N \rightarrow \mathbb{CP}^{m-1}$ is the quotient projection. ω_{FS} is nondegenerate, closed, and is in fact a Kähler form. This whole construction is the simplest nontrivial case of **symplectic reduction** (Marsden–Weinstein), with the moment map being $\mu(z) = \frac{1}{2} \|z\|^2$ for the S^1 -action.

Start concrete. $\mathbb{C}^m$ is a complex vector space of dimension m . Throw away the origin, then declare two nonzero vectors **equivalent** if one is a complex scalar multiple of the other:

$$z \sim z' \iff z' = \lambda z \text{ for some } \lambda \in \mathbb{C}^*.$$

The set of equivalence classes is **complex projective space** $\mathbb{CP}^{m-1}$. A point of $\mathbb{CP}^{m-1}$ is therefore a *complex line through the origin* in $\mathbb{C}^m$. We write the equivalence class of $(z_0, z_1, \dots, z_{m-1})$ as

$$[z_0 : z_1 : \cdots : z_{m-1}],$$

with colons reminding you that only ratios matter.

There are two equivalent ways to package this:

- **Lines through 0 picture.** Each point of $\mathbb{CP}^{m-1}$ is a complex line.
- **Sphere quotient picture.** Each line meets the unit sphere $S^{2m-1} \subset \mathbb{C}^m$ in a circle (multiply by any $e^{i\theta}$ and you stay on the line and on the sphere). So

$$\mathbb{CP}^{m-1} = S^{2m-1}/S^1,$$

which is the perspective of the previous discussion.

Both pictures will appear: the chart construction below uses the "lines" view, and the section **s** picks out a unit-norm representative on each line, connecting back to the sphere.

Manifolds are studied through **charts**: local coordinate systems that identify a piece of the manifold with an open subset of Euclidean (or in this case complex) space. To do calculus on $\mathbb{CP}^{m-1}$, we need such charts.

The standard atlas. For each $k = 0, 1, \dots, m-1$, define the open set

$$U_k = \{[z_0 : \cdots : z_{m-1}] \in \mathbb{CP}^{m-1} : z_k \neq 0\}.$$

These cover $\mathbb{CP}^{m-1}$ (every point has at least one nonzero coordinate). On U_k , since $z_k \neq 0$, we can rescale the representative by dividing through by z_k :

$$[z_0 : \cdots : z_{m-1}] = \left[\frac{z_0}{z_k} : \cdots : 1 : \cdots : \frac{z_{m-1}}{z_k} \right]$$

(the 1 is in the k -th slot). The remaining $m-1$ ratios are well-defined complex numbers and give a bijection $U_k \rightarrow \mathbb{C}^{m-1}$.

The chart on U_0 . Take $k = 0$, so we look at points where $z_0 \neq 0$.

Define

$$w_j = \frac{z_j}{z_0}, j = 1, \dots, m-1.$$

The map $[z_0:z_1:\dots:z_{m-1}] \mapsto (w_1, \dots, w_{m-1})$ is a homeomorphism (in fact biholomorphism) $U_0 \rightarrow \mathbb{C}^{m-1}$. The w_j are **local complex coordinates** on $\mathbb{CP}^{m-1}$ near any point with $z_0 \neq 0$.

This is the analog of looking at the affine plane $z_0 = 1$ inside $\mathbb{C}^m$: every line through the origin not parallel to that plane meets it in exactly one point, and that point's last $m-1$ coordinates are the w_j .

Mental picture. Just as on the round sphere we use stereographic projection to flatten a piece of sphere into a plane and do calculus there, here we flatten a piece of projective space into $\mathbb{C}^{m-1}$ using the ratios w_j . The form ω_{FS} lives on $\mathbb{CP}^{m-1}$; to write it down concretely we'll express it in these flattened coordinates.

Now comes a beautiful trick. We have:

- The sphere $\mathcal{S}_N \subset \mathbb{C}^m$, with the pulled-back form $\iota^* \omega_0$.
- The projection $\pi: \mathcal{S}_N \rightarrow \mathbb{CP}^{m-1}$, $z \mapsto [z]$.
- The quotient form ω_{FS} on $\mathbb{CP}^{m-1}$, defined by $\pi^* \omega_{FS} = \iota^* \omega_0$.

We want to write ω_{FS} in our coordinates $(w_1, \dots, w_{m-1})$ on U_0 . The cleanest way: find a section of π over U_0 — that is, a smooth map

$$s: \mathbb{C}^{m-1} \rightarrow \mathcal{S}_N \subset \mathbb{C}^m$$

Constructing s : Start with the line corresponding to $w = (w_1, \dots, w_{m-1})$. A natural representative on this line, with $z_0 = 1$, is

$$(1, w_1, \dots, w_{m-1}) \in \mathbb{C}^m.$$

This vector has squared norm

$$\| (1, w_1, \dots, w_{m-1}) \|^2 = 1 + \sum_{j=1}^{m-1} |w_j|^2 = 1 + \|w\|^2.$$

To get a unit vector on the same line, divide by the norm:

$$s(w) = \frac{1}{\sqrt{1 + \|w\|^2}} (1, w_1, \dots, w_{m-1}).$$

That's the formula in the image. Note it is *not* the only choice — we could multiply by any phase $e^{i\varphi(w)}$ and still get a section. Different sections give different liftings, but the *pulled-back form* $s^*\omega_0$ will be the same regardless, because phase shifts move you along the kernel direction iz of $\iota^*\omega_0$. This is the "gauge invariance" that makes the construction well-defined.

Why use a section instead of working on $\mathbb{CP}^{m-1}$ directly? Because forms on $\mathbb{C}^m$ are easy to write — we have global coordinates $z_0, \dots, z_{m-1}$. Forms on $\mathbb{CP}^{m-1}$ are awkward to write directly. So we lift back to $\mathbb{C}^m$, compute there, and pull the answer down. The section s is the lift.

The relationship between $s^*\omega_0$, $\iota^*\omega_0$, and ω_{FS} is:

$$s^*\omega_0 = s^*(\iota^*\omega_0) \text{ (since } s \text{ lands in } \mathcal{S}_N) = (\iota \circ s)^*\omega_0,$$

and using $\pi \circ s = \text{id}_{U_0}$ and $\pi^*\omega_{FS} = \iota^*\omega_0$:

$$s^*(\iota^*\omega_0) = s^*\pi^*\omega_{FS} = (\pi \circ s)^*\omega_{FS} = \omega_{FS}|_{U_0}.$$

So computing $s^*\omega_0$ is computing ω_{FS} in our chart. That's the whole strategy.

If you have a 2-form on $\mathbb{C}^m$ written as

$$\alpha = \sum f_{kl}(z) dz_k \wedge d\bar{z}_l$$

and a smooth map $s: \mathbb{C}^{m-1} \rightarrow \mathbb{C}^m$ given by component functions $z_k = s_k(w)$, then $s^*\alpha$ is computed by the algorithm:

1. Replace each function $f_{kl}(z)$ by $f_{kl}(s(w))$.
2. Replace each dz_k by $\left(\sum_j \frac{\partial s_k}{\partial w_j} dw_j\right)$ (and similarly $d\bar{z}_l$ by $\left(\sum_j \frac{\partial \bar{s}_l}{\partial \bar{w}_j} d\bar{w}_j\right)$)
3. Wedge everything out, collect terms.

On a complex manifold the exterior derivative splits as

$d = \partial + \bar{\partial}$, where:

- $\partial f = \sum_j \frac{\partial f}{\partial z_j} dz_j$ is the holomorphic part.
- $\bar{\partial} f = \sum_j \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$ is the antiholomorphic part.

Two key identities:

- $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial \bar{\partial} = -\bar{\partial} \partial$.
- For any smooth real-valued function K , the 2-form $i \partial \bar{\partial} K$ is automatically real and closed. (Closed because $d(\partial \bar{\partial} K) = (\partial + \bar{\partial}) \partial \bar{\partial} K = 0$ using $\partial^2 = \bar{\partial}^2 = 0$.)

A function K such that the symplectic/Kähler form is $\omega = i \partial \bar{\partial} K$ is called a **Kähler potential**. The standard form on $\mathbb{C}^m$ has Kähler potential $K(z) = \frac{1}{2} \|z\|^2$ since

$$i \partial \bar{\partial} \left(\frac{1}{2} \sum_k z_k \bar{z}_k \right) = \frac{i}{2} \sum_k dz_k \wedge d\bar{z}_k = \omega_0.$$

The famous claim: On the chart U_0 of $\mathbb{CP}^{m-1}$,

$$\omega_{FS} = i \partial \bar{\partial} \log(1 + \|w\|^2),$$

where $\|w\|^2 = \sum_{j=1}^{m-1} |w_j|^2$, the idea is that we must introduce the logarithm as is we naively replaces $z = s(w)$ in the *kähler* form

$$\omega_0 = \frac{1}{2} \partial \bar{\partial} \left(\frac{1}{2} \|s(w)\|^2 \right)$$

Then nothing but zero will be shown, we want a construction that behaves nicely when we replace z by λz , Look at $\log \|z\|^2$

If we rescale $z \rightarrow \lambda z$ then

$$\log \|\lambda z\|^2 = \log(|\lambda|^2 \|z\|^2) = \log |\lambda|^2 + \log \|z\|^2$$

But the constant part vanishes upon differentiation.

Let us derive this from

$$\omega_{FS} = i \partial \bar{\partial} \log(1 + \|w\|^2)$$

Let $f(w, \bar{w}) = 1 + \sum_k w_k \bar{w}_k$. We want $i \partial \bar{\partial} \log f$.

Step 1: $\bar{\partial} \log f$. Compute:

$$\frac{\partial}{\partial \bar{w}_\ell} \log f = \frac{1}{f} \cdot \frac{\partial f}{\partial \bar{w}_\ell} = \frac{w_\ell}{f}.$$

So

$$\bar{\partial} \log f = \sum_\ell \frac{w_\ell}{f} d\bar{w}_\ell.$$

Apply ∂ to the above. $\partial g = \sum_j \frac{\partial g}{\partial w_j} dw_j$, and ∂ acts on the coefficient and

treats $d\bar{w}_\ell$ as a constant 1-form (we then wedge dw_j on the *left*).

For each ℓ :

$$\frac{\partial}{\partial w_j} \left(\frac{w_\ell}{f} \right) = \frac{\delta_{j\ell}}{f} - \frac{w_\ell}{f^2} \cdot \frac{\partial f}{\partial w_j} = \frac{\delta_{j\ell}}{f} - \frac{w_\ell \bar{w}_j}{f^2}.$$

Therefore

$$\partial \bar{\partial} \log f = \sum_{j,\ell} \left(\frac{\delta_{j\ell}}{f} - \frac{w_\ell \bar{w}_j}{f^2} \right) dw_j \wedge d\bar{w}_\ell.$$

Combine the fractions over the common denominator f^2 :

$$\frac{\delta_{j\ell}}{f} - \frac{w_\ell \bar{w}_j}{f^2} = \frac{f \delta_{j\ell} - \bar{w}_j w_\ell}{f^2} = \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2}.$$

Step 3: Multiply by i .

$$\omega_{FS} = i \sum_{j,\ell} \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2} dw_j \wedge d\bar{w}_\ell.$$

The coefficient matrix $g_{j\bar{\ell}} = \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2}$ is the **Fubini–Study Hermitian metric**.

It's positive definite (you can check), making $\mathbb{CP}^{m-1}$ into a Kähler manifold. The " $-\bar{w}_j w_\ell$ " piece is the rank-one correction that captures the fact that radial directions in w -space have already been quotiented out.

let us see how the integration of this two form would be played:

important caveat: this is a two form and $\mathbb{CP}^{m-1}$ has real dimension $2(m-1)$.

So, we require that $m=2$.

For $\mathbb{CP}^1$, the chart coordinate is just one complex number:

$$w \in \mathbb{C}$$

So

$$\omega_{FS} = \frac{idw \wedge d\bar{w}}{(1 + |w|^2)^2}$$

$$\int_{\mathbb{CP}^1} \omega_{FS} = \int_{\mathbb{C}} \frac{idw \wedge d\bar{w}}{(1 + |w|^2)^2} = \int_{\mathbb{R}^2} \frac{2}{(1 + x^2 + y^2)^2} dx dy = 2\pi$$

Take local coordinates $(x^1, \dots, x^n)$. Then the coordinate vector fields

$$\partial_1, \dots, \partial_n \text{ where } \partial_i = \frac{\partial}{\partial x^i}$$

form a basis of the tangent space.

So a vector field looks like

$$X = \sum_{i=1}^n X^i \partial_i.$$

This means that at each point, X is just a linear combination of the coordinate directions, with coefficients X^i .

Form in coordinates

The dual basis is

$$dx^1, \dots, dx^n,$$

characterized by

$$dx^i(\partial_j) = \delta_j^i.$$

So dx^i picks out the i -th component of a vector.

A 1-form looks like

$$\alpha = \sum_{i=1}^n a_i dx^i,$$

and when you feed in the vector field $X = \sum X^i \partial_i$, you get

$$\alpha(X) = \sum_{i=1}^n a_i X^i.$$

So a 1-form is like a coordinate-dependent linear functional acting on vectors.

Higher forms

A 2-form is something that eats two vectors and is alternating. In coordinates, it looks like

$$\omega = \sum_{i < j} \omega_{ij} dx^i \wedge dx^j.$$

An n -form is top-degree, so in local coordinates it has the form

$$\omega = f dx^1 \wedge \dots \wedge dx^n$$

for some scalar function f .

That means an n -form is basically a coordinate-dependent volume-measuring object.

What $i_X \omega$ does

Now let

$$X = \sum_{i=1}^n X^i \partial_i \text{ and}$$

$$\omega = f dx^1 \wedge \cdots \wedge dx^n.$$

Then

$$i_X \omega$$

is an $(n-1)$ -form obtained by inserting X into the first slot of ω .

In coordinates, the formula is

$$i_X(dx^1 \wedge \cdots \wedge dx^n) = \sum_{r=1}^n (-1)^{r-1} X^r dx^1 \wedge \cdots \wedge \widehat{dx^r} \wedge \cdots \wedge dx^n,$$

where the hat means “omit that term.”

So for the general n -form,

$$i_X \omega = f \sum_{r=1}^n (-1)^{r-1} X^r dx^1 \wedge \cdots \wedge \widehat{dx^r} \wedge \cdots \wedge dx^n.$$

This is the coordinate formula for contraction with a top form.

Lie derivative

The Lie derivative $\mathcal{L}_X$ means: “how does an object change when I drag it along the flow generated by X ?”

For a function f , this is nothing mysterious:

$$\mathcal{L}_X f = X(f),$$

which is just the directional derivative of f in the direction X

For a differential form ω , the Lie derivative measures how ω changes under the flow of X . Cartan's formula says

$$\mathcal{L}_X \omega = d(i_X \omega) + i_X(d\omega).$$

This formula is important because it turns “change along a flow” into operations on forms that we can compute with.

Divergence tells you whether a vector field is acting like a local source or sink of volume. If tiny blobs of volume expand under the flow, divergence is positive; if they shrink, divergence is negative.

On a manifold with a volume form ω , divergence is defined by

$$\mathcal{L}_X \omega = (\operatorname{div} X) \omega.$$

So divergence is the scalar coefficient describing how the volume form changes under the flow of X .

Now apply Cartan's formula to the volume form:

$$\mathcal{L}_X dV_g = d(i_X dV_g) + i_X(d dV_g).$$

But dV_g is a top-degree form, so $d(dV_g) = 0$. Therefore

$$\mathcal{L}_X dV_g = d(i_X dV_g).$$

Since the left-hand side is also $(\operatorname{div} X)dV_g$, you get

$$(\operatorname{div} X)dV_g = d(i_X dV_g).$$

This is the central identity behind the divergence theorem.

Then Stokes

Stokes' theorem says

$$\int_M d\eta = \int_{\partial M} \eta$$

for an $(n-1)$ -form η .

Now choose

$$\eta = i_X dV_g.$$

Then

$$\int_M (\operatorname{div} X) dV_g = \int_M d(i_X dV_g) = \int_{\partial M} i_X dV_g.$$

That is already the divergence theorem in form language.

Finally, on the boundary, $i_X dV_g$ becomes the flux term $g(X, \nu) dS_g$, where ν is the outward unit normal. So

$$\int_M (\operatorname{div} X) dV_g = \int_{\partial M} g(X, \nu) dS_g.$$

That is the Riemannian divergence theorem.

The exterior derivative is an operator

$$d: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

It generalizes gradient, curl, and divergence. It satisfies key properties On functions $f \in \Omega^0(M) = C^\infty(M)$

$$df = \sum_i \frac{\partial}{\partial x^i} dx^i$$

It satisfies the graded product rule

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$$

And it squares to zero ($d^2 = 0$)

In coordinates if

$$\begin{aligned} \omega &= \sum_I \omega_I dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ d\omega &= \sum_I d\omega_I dx^{i_1} \wedge \dots \wedge dx^{i_k}, \quad d\omega_I = \sum_j \frac{\partial \omega_I}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ d\omega &= \sum_{I,j} \frac{\partial \omega_I}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} \end{aligned}$$

One important identity is that pullback commutes with the exterior derivative

$$d(F^* \omega) = F^*(d\omega)$$

We'll be using this fact to compute $d\omega$ before or after changing coordinates without worrying about consistency.

Let M be an oriented n -manifold. Suppose $\omega \in \Omega^n(M)$. In a positively oriented coordinate chart,

$$\omega = f(x^1, \dots, x^n) dx^1 \wedge \dots \wedge dx^n$$

Then locally we define

$$\int_U \omega = \int_{\varphi(U)} f(x^1, \dots, x^n) dx^1 \dots dx^n$$

This is coordinate-independent because under a change of variables, the top form transforms by the determinant of the Jacobian.

The generalized Stokes theorem says:

$$\int_M d\alpha = \int_{\partial M} \alpha$$

If the coordinate change is orientation-preserving, the determinant is positive, for a general manifold, one uses a partition of unity to glue the local definitions together.

For a smooth function f , df is a one form. The gradient ∇f is defined by

$$g(\nabla f, X) = df(X)$$

For all vector fields X . *df exists without a metric, but ∇f requires a metric.* The Laplacian $\Delta f = \text{div}(\nabla f)$ requires metric.

Here M is an oriented n -manifold with boundary, and

$$\alpha \in \Omega^{n-1}(M)$$

A manifold M is **orientable** if you can choose, at each point p , an equivalence class of ordered bases of $T_p M$ (two bases are equivalent if the change-of-basis matrix has positive determinant), in a way that varies continuously across M .

Three equivalent ways to package this:

1. **Pointwise:** a continuous choice of "positive frame" at each tangent space.
2. **Form-theoretic:** a nowhere-vanishing top-form $\Omega \in \Omega^n(M)$. Two such forms give the same orientation iff they differ by a positive function.
3. **Atlas-theoretic:** an atlas where every chart transition map has positive Jacobian determinant.

is the space of complex lines through the origin in $\mathbb{C}^3$, points written as $[z_0:z_1:z_2]$. Complex dimension 2, real dimension 4.

Three charts, the standard atlas. For $k = 0,1,2$, the open set

$$U_k = \{[z_0:z_1:z_2]: z_k \neq 0\}.$$

These cover $\mathbb{CP}^2$. Local complex coordinates:

- On U_0 : $w_1 = z_1/z_0$, $w_2 = z_2/z_0$. So $[z_0:z_1:z_2] = [1:w_1:w_2]$.
- On U_1 : $\zeta_0 = z_0/z_1$, $\zeta_2 = z_2/z_1$. So $[z_0:z_1:z_2] = [\zeta_0:1:\zeta_2]$.
- On U_2 : $\eta_0 = z_0/z_2$, $\eta_1 = z_1/z_2$.

Each chart maps its open set to $\mathbb{C}^2$, which we identify with $\mathbb{R}^4$ via $w_1 = x_1 + iy_1$, $w_2 = x_2 + iy_2$. So real coordinates on U_0 are (x_1, y_1, x_2, y_2) .

Transition map $U_0 \cap U_1$. Where both $z_0, z_1 \neq 0$ — i.e., $w_1 = z_1/z_0 \neq 0$. Compute:

$$\zeta_0 = \frac{z_0}{z_1} = \frac{1}{w_1}, \zeta_2 = \frac{z_2}{z_1} = \frac{w_2}{w_1}.$$

This is a *holomorphic* map $\Phi: \{w_1 \neq 0\} \rightarrow \mathbb{C}^2$. Hold on to this — it's the heart of the calculation.

Pick a point $p \in U_0$. Its tangent space $T_p \mathbb{CP}^2$ is real 4-dimensional. The complex coordinates w_1, w_2 give the *real* coordinate frame

$$(\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2}) \in T_p \mathbb{CP}^2.$$

Choose the orientation $[\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2}]$.

That's a pointwise orientation at p : the equivalence class of *that* ordered basis. Why this ordering? Because it's the "complex" ordering: real, imaginary, real, imaginary. The complex structure J (multiplication by i) acts as $J \partial_{x_j} = \partial_{y_j}$ and $J \partial_{y_j} = -\partial_{x_j}$, and the chosen ordering is compatible with J in a specific sense we'll make precise in step 5.

Continuity. Vary p within U_0 : the frame $(\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2})$ is the coordinate frame, smooth in p , so the chosen pointwise orientation extends continuously across U_0 .

Now extend to U_1 . Pick $q \in U_0 \cap U_1$. Two candidate frames at q :

- From U_0 : $F_0 = (\partial_{x_1}, \partial_{y_1}, \partial_{x_2}, \partial_{y_2})$ (in w -coords).
- From U_1 : $F_1 = (\partial_{u_0}, \partial_{v_0}, \partial_{u_2}, \partial_{v_2})$, where $\zeta_0 = u_0 + iv_0$, $\zeta_2 = u_2 + iv_2$.

Are these in the same equivalence class? That's a determinant computation.

Differentiating the transition Φ :

$$\zeta_0 = \frac{1}{w_1}, \zeta_2 = \frac{w_2}{w_1} \Rightarrow \frac{\partial \zeta_0}{\partial w_1} = -\frac{1}{w_1^2}, \frac{\partial \zeta_0}{\partial w_2} = 0, \frac{\partial \zeta_2}{\partial w_1} = -\frac{w_2}{w_1^2}, \frac{\partial \zeta_2}{\partial w_2} = \frac{1}{w_1}.$$

The complex Jacobian matrix is

$$J_{\mathbb{C}} = \begin{pmatrix} -1/w_1^2 & 0 \\ -w_2/w_1^2 & 1/w_1 \end{pmatrix}, \det J_{\mathbb{C}} = -\frac{1}{w_1^3}.$$

A *complex* number, possibly negative, possibly far from positive. So you might worry. Don't! What matters is the *real* Jacobian, the 4×4 matrix of partial derivatives of (u_0, v_0, u_2, v_2) with respect to (x_1, y_1, x_2, y_2) .

Key lemma. If $f: \mathbb{C}^n \rightarrow \mathbb{C}^n$ is holomorphic with complex Jacobian $J_{\mathbb{C}}$, then its real Jacobian (as a map $\mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$) satisfies

$$\det J_{\mathbb{R}} = |\det J_{\mathbb{C}}|^2 \geq 0$$

When you integrate an n -form ω over M , you do it chart by chart and sum. In each chart, $\int_U \omega$ is computed as an ordinary Lebesgue integral of the coefficient function. If the chart transitions had negative Jacobians, two adjacent charts would disagree on the sign of the same piece of integral, and the sum would be ill-defined. Orientation is exactly the structure that prevents this.

A **manifold with boundary** M is locally modeled on the half-space $\mathbb{H}^n = \{x \in \mathbb{R}^n: x^n \geq 0\}$. The boundary ∂M is an $(n-1)$ -manifold without boundary.

The induced orientation rule. If M is oriented, the boundary inherits a canonical orientation by the **outward-normal-first rule**: a basis $(v_1, \dots, v_{n-1})$ of $T_p \partial M$ is positively oriented iff $(v, v_1, \dots, v_{n-1})$ is positively oriented in $T_p M$, where v is an outward-pointing vector. This is a convention, but it's the one that makes Stokes work without an extra sign.

1.4 Integration of forms — the mechanism

The actual definition is engineered so that the result is independent of all choices.

Step 1 — In a single oriented chart. If $\omega = f dx^1 \wedge \dots \wedge dx^n$ is supported in an oriented chart $U \subset \mathbb{R}^n$,

$$\int_M \omega := \int_U f(x) dx^1 \dots dx^n,$$

an ordinary Lebesgue integral. The ordering matches the orientation.

Step 2 — Partition of unity. For a general ω on M , pick a locally finite oriented atlas $\{U_\alpha\}$ and a partition of unity $\{\rho_\alpha\}$ subordinate to it. Then

$$\int_M \omega := \sum_\alpha \int_{U_\alpha} \rho_\alpha \omega.$$

Step 3 — Independence of choices. One proves this is independent of the atlas and partition by showing that on overlaps the change-of-variables formula plus the wedge-product transformation rule cancel exactly. Orientability is exactly the hypothesis that makes the cancellation produce the *same* sign in every chart.

How to think about partitions of unity. They are not a deep technical trick; they are a smooth book-keeping device. The functions ρ_α are smooth, nonnegative, locally finitely many, sum to 1, and each is supported in one chart. Multiplying any global object by ρ_α localizes it to a chart, where you can compute. The sum then reassembles a global answer. They exist on any paracompact manifold, which includes essentially every manifold you ever meet.

So far, forms do not require a metric. A smooth manifold gives you

$$T_p M, \quad T_p^* M, \quad \Omega^k(M), d, \quad \wedge, \quad F^*$$

But it does not automatically give you lengths, angles gradients, or volume. A Riemannian metric is a smoothly varying inner product:

Definition (Riemannian metric). Let M be a smooth n -manifold. A Riemannian metric on M is a smooth assignment of an inner product $g_p: T_p M \times T_p M \rightarrow \mathbb{R}$ to every point $p \in M$, such that in any chart (U, φ) with coordinates $(x_1, \dots, x_n)$, the matrix $g_{ij}(x) = g_p(\partial_i, \partial_j)$

is symmetric and positive definite, and the entries $g_{ij}(x)$ depend smoothly on x . The pair (M, g) is called a Riemannian manifold. We write g^{ij} for the inverse matrix; it represents the induced inner product on the cotangent space $T_p^* M$.

Example (The standard Hermitian / Riemannian structure on $\mathbb{C}^m$). Identify $\mathbb{C}^m \simeq \mathbb{R}^{2m}$ via $z_k = a_k + ib_k$. The standard Hermitian inner product $\langle z, w \rangle_{\mathbb{C}^m} = \sum_k \bar{z}_k w_k$ decomposes into real and imaginary parts

$$\langle z, w \rangle_{\mathbb{C}^m} = g(z, w) + i\omega_0(z, w), \quad (2)$$

where $g(z, w) = \text{Re} \langle z, w \rangle$ is the standard Euclidean metric on $\mathbb{R}^{2m}$, and $\omega_0(z, w) = \text{Im} \langle z, w \rangle$ is the standard symplectic form. Thus, on $\mathbb{C}^m$ the metric and the symplectic form are real and imaginary halves of one Hermitian object. This decomposition is the foundation of Kähler geometry and reappears throughout the worked examples.

A **symplectic form** is a closed, non-degenerate, antisymmetric 2-form on a manifold. It's the geometric object that lets you do **Hamiltonian mechanics** — it tells you how positions and momenta are paired.

A Riemannian metric g measures lengths and angles. A symplectic form ω measures **signed areas**.

In classical mechanics, phase space T^*M (positions + momenta) carries a canonical symplectic form $\omega = \sum dp_i \wedge dq_i$. Given a Hamiltonian H , the equations of motion are

$$\omega(X_H, \cdot) = dH,$$

which unpacks to Hamilton's equations $\dot{q}_i = \partial H / \partial p_i$, $\dot{p}_i = -\partial H / \partial q_i$. The symplectic form is the geometric content of " p and q are conjugate variables."

Let (M, g) be an oriented Riemannian n manifold. The Riemannian volume form $dvolg$ is the unique smooth n -form on M such that on any positively oriented g -orthonormal frame $(e_1, \dots, e_n)$ of $T_p M$, $dvolg(e_1, \dots, e_n) = 1$

Let V be an n -dimensional vector space and $\omega \in \Lambda^n V^*$ a top form (i.e., an alternating n -multilinear functional). If $(e_1, \dots, e_n)$ and $(f_1, \dots, f_n)$ are two bases related by $f_a = M_a^i e_i$, then

$$\omega(f_1, \dots, f_n) = (\det M) \omega(e_1, \dots, e_n).$$

This is just the multilinearity-and-antisymmetric of ω :

expanding $\omega(M_1^{i_1} e_{i_1}, \dots, M_n^{i_n} e_{i_n})$ and using antisymmetry to collect terms produces exactly the determinant.

Let $A = (A_a^i)$ be the matrix sending the coordinate frame $(\partial_1, \dots, \partial_n)$ to a positively oriented g -orthonormal frame: $ea = A^i_a \partial_i$. Orthonormality reads

$$\delta_{ab} = g(ea, eb) = A_a^i A_b^j g_{ij}, \quad I = A^T G A \quad - \quad (5)$$

where $G = (g_{ij})$. Taking determinants gives $(\det A)^{-1} = (\det G)$. Two top forms are related by the determinant of a change of basis, so evaluating $dvolg$ on the coordinate frame yields

$$dvolg(\partial_1, \dots, \partial_n) = (\det A)^{-1} = \sqrt{\det G}$$

proving the coordinate formula.

The qutrit example $\mathbb{CP}^2$ (real dimension 4) is the perfect playground: a single chart misses a complex projective line's worth of states, so you genuinely cannot compute the volume from one chart — you *must* glue charts together with a partition of unity.

Worked example: Fubini–Study volume of $\mathbb{CP}^2$ (the qutrit state space) via partition of unity

What's the manifold?

A pure qutrit state is a vector $[\alpha_0 : \alpha_1 : \alpha_2] \in \mathbb{C}^3$ modulo overall complex scaling. The state space is

$$\mathbb{CP}^2 = \{[\alpha_0 : \alpha_1 : \alpha_2]\}$$

— a compact, real 4-dimensional manifold (4 real coordinates because we kill one complex degree of freedom: phase + normalisation). The Fubini–Study metric is the natural quantum-mechanical metric: $g_{\text{FS}}(\dot{\psi}, \dot{\psi})$ measures the rate at which a state becomes distinguishable from itself under unitary evolution.

Why one chart cannot suffice

Any standard affine chart on $\mathbb{CP}^2$ is built by setting one of the homogeneous coordinates equal to 1. There are three of them:

- $U_0 = \{\alpha_0 \neq 0\}$, with chart $\varphi_0([\alpha]) = (\alpha_1/\alpha_0, \alpha_2/\alpha_0) \in \mathbb{C}^2$
- $U_1 = \{\alpha_1 \neq 0\}$, with chart $\varphi_1([\alpha]) = (\alpha_0/\alpha_1, \alpha_2/\alpha_1) \in \mathbb{C}^2$
- $U_2 = \{\alpha_2 \neq 0\}$, with chart $\varphi_2([\alpha]) = (\alpha_0/\alpha_2, \alpha_1/\alpha_2) \in \mathbb{C}^2$

Each chart misses a "hyperplane at infinity" — a copy of $\mathbb{CP}^1$. For example, U_0 misses the qubit $\{[0:\alpha_1:\alpha_2]\} \cong \mathbb{CP}^1$. That's a 2-dimensional submanifold — *not* measure zero in any natural sense for a single chart, even though it has measure zero in $\mathbb{CP}^2$ itself.

Conceptually you can get away with a single chart for the volume of $\mathbb{CP}^1$ because what's missing is a single point. For $\mathbb{CP}^2$ what's missing is a whole sphere. So we need the partition-of-unity machinery to handle this honestly.

Set $z = (z_1, z_2) = (\alpha_1/\alpha_0, \alpha_2/\alpha_0)$. The Fubini–Study line element is

$$ds^2 = \frac{(1+|z|^2) |dz|^2 - |\bar{z} \cdot dz|^2}{(1+|z|^2)^2}, \quad |z|^2 = |z_1|^2 + |z_2|^2.$$

In real coordinates $z_k = x_k + iy_k$ for $k = 1, 2$, this becomes a 4×4 symmetric positive-definite matrix g_{ij} on $\mathbb{R}^4$. It looks ugly, but its **determinant** is famously clean:

$$\det g_{ij}(z) = \frac{1}{(1+|z|^2)^6}.$$

So by the coordinate volume formula,

$$d\text{vol}_{\text{FS}} = \sqrt{\det g_{ij}} \, dx_1 \, dy_1 \, dx_2 \, dy_2 = \frac{1}{(1+|z|^2)^3} \, dx_1 \, dy_1 \, dx_2 \, dy_2.$$

By symmetry (the metric is built to be invariant under the unitary group $U(3)$, which permutes the charts U_0, U_1, U_2), the same density formula holds in U_1 and U_2 with their respective coordinates.

Here is a clean choice. For $[\alpha] = [\alpha_0 : \alpha_1 : \alpha_2]$ with $\sum |\alpha_k|^2 = 1$ (i.e., a unit representative), set

$$\rho_i([\alpha]) = \frac{|\alpha_i|^2}{|\alpha_0|^2 + |\alpha_1|^2 + |\alpha_2|^2} = |\alpha_i|^2.$$

These satisfy $\rho_0 + \rho_1 + \rho_2 = 1$ trivially. Each ρ_i vanishes precisely where its chart fails ($\alpha_i = 0$), and is well-defined on $\mathbb{CP}^2$ (the ratio is invariant under $[\alpha] \mapsto [\lambda\alpha]$). They're not compactly supported in the chart in the strict topological sense — they decay at infinity instead — but they're integrable, which is what equation (7) actually needs. Strictly speaking we'd smooth them out to be compactly supported; the integrals are unchanged.

In chart U_0 coordinates $z = (z_1, z_2)$:

$$\rho_0 = \frac{1}{1 + |z|^2}, \rho_1 = \frac{|z_1|^2}{1 + |z|^2}, \rho_2 = \frac{|z_2|^2}{1 + |z|^2}.$$

The total volume is

$$\text{Vol}(\mathbb{CP}^2) = \int_{\mathbb{CP}^2} 1 \cdot d\text{vol}_{\text{FS}} = \sum_{i=0}^2 \int_{\mathbb{CP}^2} \rho_i d\text{vol}_{\text{FS}}.$$

Each piece ρ_i is supported in U_i , so the i -th integral pulls back to a chart integral via equation (7):

$$\int_{\mathbb{CP}^2} \rho_i d\text{vol}_{\text{FS}} = \int_{\varphi_i(U_i)} \rho_i(z) \sqrt{\det g_{ij}(z)} dx_1 dy_1 dx_2 dy_2.$$

This is *exactly* equation (7) in the screenshot — $f = \rho_i$, and we sum the three resulting honest $\mathbb{R}^4$ integrals. No double-counting because of $\sum \rho_i = 1$; no missing pieces because each U_i is missed by only one of the others.

Switch to 4D spherical coordinates: $r^2 = x_1^2 + y_1^2 + x_2^2 + y_2^2$, with surface area of the unit S^3 equal to $2\pi^2$ and volume element $d^4z = 2\pi^2 r^3 dr$:

$$\int_{\mathbb{R}^4} \frac{d^4z}{(1+|z|^2)^4} = 2\pi^2 \int_0^\infty \frac{r^3 dr}{(1+r^2)^4}.$$

Substitute $u = 1 + r^2$, $du = 2r dr$, $r^2 = u - 1$:

$$\int_0^\infty \frac{r^3 dr}{(1+r^2)^4} = \frac{1}{2} \int_1^\infty \frac{u-1}{u^4} du = \frac{1}{2} \int_1^\infty (u^{-3} - u^{-4}) du = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{12}.$$

So one chart's contribution is $2\pi^2 \cdot \frac{1}{12} = \frac{\pi^2}{6}$, and

$$\text{Vol}(\mathbb{CP}^2) = 3 \cdot \frac{\pi^2}{6} = \boxed{\frac{\pi^2}{2}}.$$

Two operators distinguish a Riemannian manifold from a bare smooth one. Each requires the metric in an essential way

Let $f \in C^\infty(M)$. The gradient of f is the unique vector field ∇f such that $g(\nabla f, X) = df(X) = X(f)$ for every vector field X . In coordinates, $(\nabla f)_i = g_{ij} \partial_j f$.

The differential df is a 1-form and exists on any smooth manifold. The gradient ∇f is a vector field, obtained from df by raising the index using g^{ij} . Without a metric, there is no canonical way to turn a covector into a vector, so there is no canonical gradient. This is the content of the often-quoted slogan “ ∇f uses a metric, df does not.”

Let X be a smooth vector field on (M, g) . The divergence of X is the unique smooth function $\text{div}_g X$ such that

$$\mathcal{L}_X(d\text{vol}_g) = (\text{div}_g X) d\text{vol}_g$$

Geometrically, $\text{div}_g X$ measures the infinitesimal rate at which the Riemannian volume form is stretched by the flow of X . Cartan's magic formula

$$\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$$

together with the identity $d(d\text{vol}_g) = 0$ (because $d\text{vol}_g$ is a top form and so its exterior derivative is automatically zero) gives the equivalent expression

$$(\text{div}_g X) d\text{vol}_g = d(\iota_X d\text{vol}_g)$$

Stokes's theorem applied to the $(n - 1)$ -form $\iota_X d\text{vol}_g$ gives the divergence theorem: for X compactly supported on a manifold with boundary,

$$\int_M (\text{div}_g X) d\text{vol}_g = \int_{\partial M} g(X, \nu) d\sigma$$

where the outward unit is normal and $d\sigma$ is the induced volume form on the boundary. On a closed manifold (no boundary) or for compactly supported X on a manifold without boundary, the right-hand side is zero

Let $u, v \in C_c^\infty(M)$ on (M, g) . Then

$$\int_M u \Delta_g v$$

$\Delta_g f := \text{div}_g(\nabla f)$ is the Laplace–Beltrami operator. $\int_M v \Delta_g u d\text{vol}_g$,

EXAMPLE:

Consider a spinless particle of mass m constrained to move on a Riemannian manifold (M, g) ; for definiteness take $M = S^2$ with the usual round metric of radius R .

- Coordinates: use spherical angles (θ, ϕ) with $0 < \theta < \pi, 0 < \phi < 2\pi$.
- Metric on S^2 :

$$g_{\theta\theta} = R^2, g_{\phi\phi} = R^2 \sin^2 \theta, g_{\theta\phi} = 0.$$

So

$$ds^2 = R^2(d\theta^2 + \sin^2 \theta d\phi^2).$$

Your physical Hilbert space is

$$\mathcal{H} = L^2(M, d\text{vol}_g),$$

square-integrable complex functions on M with respect to the Riemannian volume form.

For a Riemannian metric, the volume form is

$$d\text{vol}_g = \sqrt{\det g} \, d^n x$$

In our coordinates,

$$\det g = R^4 \sin^2 \theta, \sqrt{\det g} = R^2 \sin \theta,$$

so

$$d\text{vol}_g = R^2 \sin \theta \, d\theta \, d\phi.$$

This is the measure with respect to which probabilities are computed. The normalization condition for a wavefunction ψ is

$$\int_{S^2} |\psi|^2 \, d\text{vol}_g = 1,$$

which concretely is

$$\int_0^\pi \int_0^{2\pi} |\psi(\theta, \phi)|^2 R^2 \sin \theta \, d\theta \, d\phi = 1.$$

So the non-flat metric shows up directly in what “uniform probability” means on the sphere. Again, On a Riemannian manifold, the gradient of a scalar function f is defined by

$$g(\nabla f, X) = X(f) \text{ for all vector fields } X.$$

In local coordinates this becomes

$$(\nabla f)^i = g^{ij} \partial_j f,$$

- Inverse metric components are $g^{\theta\theta} = 1/R^2$, $g^{\phi\phi} = 1/(R^2 \sin^2 \theta)$, $g^{\theta\phi} = 0$.
- So, for a scalar $f(\theta, \phi)$,

$$\nabla f = \frac{1}{R^2} \partial_\theta f \, \partial_\theta + \frac{1}{R^2 \sin^2 \theta} \partial_\phi f \, \partial_\phi.$$

For a potential $V(\theta, \phi)$, $-\nabla V$ plays the role of force (in classical mechanics) on the curved surface. In quantum mechanics, ∇ is needed to build the kinetic term and to write current densities.

Short explanation of the Lie derivative

Given a vector field X on M , it generates a flow Φ_t (think: move every point along the integral curves of X for time t).

- For any tensor field T (scalar, vector, form, metric, volume form...), you can pull it back by Φ_t .
- The Lie derivative $\mathcal{L}_X T$ is the infinitesimal change of T under this flow:

$$\mathcal{L}_X T = \frac{d}{dt} \Big|_{t=0} \Phi_t^* T. \quad \square \square \square \square$$

For the volume form, $\mathcal{L}_X(d\text{vol}_g)$ measures how the flow generated by X locally stretches or compresses volume elements. This is what divergence is

From the definition via Lie derivative and Cartan's formula (as your text states), one can derive the coordinate expression

$$\text{div}_g X = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i). \quad \square \square$$

Let us do this for the sphere. Write

$$X = X^\theta(\theta, \phi) \partial_\theta + X^\phi(\theta, \phi) \partial_\phi.$$

Then

$$\text{div}_g X = \frac{1}{R^2 \sin \theta} [\partial_\theta (R^2 \sin \theta X^\theta) + \partial_\phi (R^2 \sin \theta X^\phi)].$$

Simplify the R^2 :

$$\text{div}_g X = \frac{1}{\sin \theta} [\partial_\theta (\sin \theta X^\theta) + \partial_\phi (\sin \theta X^\phi)].$$

Suppose X is a probability current or velocity field on the surface. Then $\text{div}_g X > 0$ at a point means probability is “spreading out” there; $\text{div}_g X < 0$ means it is “converging.”

This is the curved-space generalization of $\nabla \cdot \vec{j}$.

By definition, the Laplace–Beltrami operator is the divergence of the gradient:

$$\Delta_g f = \text{div}_g (\nabla f).$$

In coordinates,

$$\Delta_g f = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} g^{ij} \partial_j f).$$

We already know g^{ij} and $\sqrt{\det g} = R^2 \sin \theta$. Thus

$$\Delta_g f = \frac{1}{R^2 \sin \theta} [\partial_\theta (R^2 \sin \theta g^{\theta\theta} \partial_\theta f) + \partial_\phi (R^2 \sin \theta g^{\phi\phi} \partial_\phi f)].$$

Insert $g^{\theta\theta} = 1/R^2$, $g^{\phi\phi} = 1/(R^2 \sin^2 \theta)$:

$$\Delta_g f = \frac{1}{R^2 \sin \theta} \left[\partial_\theta (\sin \theta \partial_\theta f) + \partial_\phi \left(\frac{1}{\sin \theta} \partial_\phi f \right) \right].$$

Thus

$$\Delta_{S^2} f = \frac{1}{R^2} \left[\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta f) + \frac{1}{\sin^2 \theta} \partial_\phi^2 f \right],$$

which is the standard spherical Laplacian.

The “Gradient and divergence” definition leads directly to this formula when you apply the proposition

$$\operatorname{div}_g X = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i); X = \nabla f$$

For a free particle of mass m on (M, g) , the natural Hamiltonian is $\hat{H} = -\frac{\hbar^2}{2m} \Delta_g$.

Then the Schrödinger equation reads

$$i\hbar \frac{\partial}{\partial t} \psi(t, x) = -\frac{\hbar^2}{2m} \Delta_g \psi(t, x).$$

This form is standard in quantum mechanics on Riemannian manifolds and in the “geometrization of the Schrödinger equation” literature

Using the Laplace–Beltrami we derived, the Hamiltonian becomes

$$\hat{H}\psi = -\frac{\hbar^2}{2mR^2} \left[\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta \psi) + \frac{1}{\sin^2 \theta} \partial_\phi^2 \psi \right].$$

On (M, g) , the probability current associated with ψ is

$$J^i = \frac{\hbar}{2mi} (\psi^* \nabla^i \psi - \psi \nabla^i \psi^*),$$

and the continuity equation is

$$\frac{\partial}{\partial t} |\psi|^2 + \operatorname{div}_g J = 0.$$

NUMERICAL ANALYSIS:

PART I – FRECHET DERIVATIVES IN FOURIER SPACE:

Let $\mathcal{B} = \ell^2(\mathbb{Z}, \mathbb{C})$ be the Hilbert space of square-summable bilateral complex sequences $\hat{f} = (\hat{f}_k)_{k \in \mathbb{Z}}$ with norm

$$\|\hat{f}\|_{\ell^2} = \left(\sum_{k \in \mathbb{Z}} |\hat{f}_k|^2 \right)^{\frac{1}{2}}$$

Essentially what we're doing here is sampling a function in frequency space, by finding its projections onto the continuous Fourier basis:

$$\hat{f}_k = \langle f, e^{ikx} \rangle_{L^2(\mathbb{T})} = \int_{\mathbb{T}} f(x) e^{-ikx} dx$$

We Identify $\hat{f}$ with the $L^2(\mathbb{T})$ function: the inverse Fourier transform $f(x) = \sum_{k \in \mathbb{Z}} \hat{f}_k e^{ikx}$

Define the Fourier shift operator $S_h: \mathcal{B} \rightarrow \mathcal{B}$ by

$$(S_h \hat{f})_k = e^{ikh} \hat{f}_k, \quad h \in \mathbb{R},$$

Which corresponds to the spatial translation $(S_h f)(x) = f(x + h)$ in physical space.

Exercise 1 — Fourier Forward Difference as a Bounded Operator.

$$\Delta_h^{\mathcal{F}}: \mathcal{B} \rightarrow \mathcal{B}, \Delta_h^{\mathcal{F}} \hat{f} = S_h \hat{f} - \hat{f}, \quad i.e., \quad (\Delta_h^{\mathcal{F}} \hat{f})_k = (e^{ikh} - 1) \hat{f}_k$$

Part(a) — Boundedness and operator Norm

$$\begin{aligned} \Delta_h^{\mathcal{F}}(\hat{f} + \alpha \hat{g}) &= \left(\Delta_h^{\mathcal{F}}(\hat{f} + \alpha \hat{g}) \right)_k = (e^{ikh} - 1)(\hat{f}_k + \alpha \hat{g}_k) = (e^{ikh} - 1)\hat{f}_k + \alpha(e^{ikh} - 1)\hat{g}_k \\ &= (\Delta_h^{\mathcal{F}} \hat{f})_k + \alpha(\Delta_h^{\mathcal{F}} \hat{g})_k = \Delta_h^{\mathcal{F}} \hat{f} + \alpha \Delta_h^{\mathcal{F}} \hat{g} \end{aligned}$$

$$\|\Delta_h^{\mathcal{F}} \hat{f}\|_{\ell^2} = \left(\sum_{k \in \mathbb{Z}} (|e^{ikh} - 1|)^2 |\hat{f}_k|^2 \right)^{\frac{1}{2}} \leq \left(\sum_{k \in \mathbb{Z}} K^2 |\hat{f}_k|^2 \right)^{\frac{1}{2}} = K \|\hat{f}\|_{\ell^2}$$

$$\|\Delta_h^{\mathcal{F}} \hat{f}\|_{\ell^2} \leq K \|\hat{f}\|_{\ell^2}, \quad K = \sup_{k \in \mathbb{Z}} |e^{ikh} - 1|$$

Hence $\Delta_h^{\mathcal{F}}$, is a bounded operator,

Case 1: $h/\pi \in \mathbb{Q}$ (rational multiple)

Write $h = \frac{p}{q}\pi$ in lowest terms. Then $kh = \frac{kp}{q}\pi$, and as k ranges over $\mathbb{Z}$, the values $kh \bmod 2\pi$ cycle through a **finite set** of $2q$ equally spaced points on the circle. The supremum $|e^{ikh} - 1| = 2 |\sin(kh/2)|$ is achieved at the point in this finite set closest to π . It may or may not hit exactly π , so the sup may be **strictly less than 2** and is a **maximum**.

as $|e^{ikh} - 1|$ is a bounded function $\exists k_0 \in \mathbb{Z}; \sup_{k \in \mathbb{Z}} |e^{ikh} - 1| = |e^{ik_0 h} - 1|$ Letting $(\hat{f})_k = \delta_{k, k_0}$

$$\|\Delta_h^{\mathcal{F}} \hat{f}\|_{\ell^2} = |e^{ik_0 h} - 1| = 2 |\sin(kh/2)| = 2$$

achieved at any k such that kh is an **odd multiple** of π .

Case 2: $h/\pi \notin \mathbb{Q}$ (irrational multiple)

By **Weyl's equidistributional theorem**, the sequence $\{kh, \bmod 2\pi\}_{k \in \mathbb{Z}}$ is **dense** in $[0, 2\pi]$. There exist integers k_n such that $k_n h \rightarrow \pi \pmod{2\pi}$, so:

$$|e^{ik_n h} - 1| = 2 |\sin\left(\frac{k_n h}{2}\right)| \rightarrow 2 |\sin(\pi/2)| = 2.$$

The value 2 is **never attained** (since $k_n h = \pi$ exactly would require $h/\pi \in \mathbb{Q}$), but the supremum equals 2. So, the operator norm is still **2**, but it's a supremum, **not a maximum** — no single $\hat{f}$ of unit norm achieves it.

Part (b) — First – Order Error in Operator Norm

Define the Fourier derivative approximation operator $\Phi_h^{\mathcal{F}} = \frac{1}{h} \Delta_h^{\mathcal{F}}$, and the spectral derivative $D \in \mathcal{L}(\mathcal{B}_1, \mathcal{B})$ by $(D\hat{f})_k = ik\hat{f}_k$, which corresponds to $\partial_x f$ in physical space.

$$\|(\Phi_h^{\mathcal{F}} - D)\hat{f}\|_{\ell^2}^2 = \sum_{k \in \mathbb{Z}} \left| \frac{(e^{ikh} - 1)}{h} - ik \right|^2 |\hat{f}_k|^2$$

Applying a Taylor approximation at 0 we see:

$$e^{ikh} = 1 + ik(h) + \frac{(ik)^2}{2} h^2 e^{i\theta kh}, \quad \text{for } \theta \in [0,1]$$

Rearranging we get:

$$\begin{aligned} \frac{e^{ikh} - 1}{h} - ik &= \frac{(ik)^2 h}{2} e^{i\theta kh} = \frac{(ikh)^2}{2h} e^{i\theta kh} \\ \|(\Phi_h^{\mathcal{F}} - D)\hat{f}\|_{\ell^2}^2 &= \sum_{k \in \mathbb{Z}} \left| \frac{(e^{ikh} - 1)}{h} - ik \right|^2 |\hat{f}_k|^2 \\ &= \sum_{k \in \mathbb{Z}} \left| \frac{(ikh)^2}{2h} e^{i\theta kh} \right|^2 |\hat{f}_k|^2 = \left(\frac{h}{2}\right)^2 \sum_{k \in \mathbb{Z}} |k^2|^2 |\hat{f}_k|^2 = \left(\frac{h}{2}\right)^2 \sum_{k \in \mathbb{Z}} |-k^2 \hat{f}_k|^2 \end{aligned}$$

And therefore one observes that:

$$\|(\Phi_h^{\mathcal{F}} - D)\hat{f}\|_{\ell^2} = \left(\frac{h}{2}\right) \|D^2 \hat{f}\|_{\ell^2}$$

Part (c) — Second-Order Centered Scheme in ℓ^2

Now observe the following:

$$\begin{aligned} e^{ikh} &= 1 + ik(h) + \frac{(ik)^2}{2} h^2 + \frac{(ik)^3 h^3}{6} e^{i\theta k}, \quad \text{for } \theta \in [0,1] \\ -e^{-ikh} &= -1 + ik(h) - \frac{(ik)^2}{2} h^2 + \frac{(ik)^3 h^3}{6} e^{i\theta k}, \quad \text{for } \theta \in [0,1] \end{aligned}$$

So adding both equations yield

$$e^{ikh} - e^{-ikh} = 2ikh(h) + \frac{(ik)^3 h^3}{3} e^{i\theta k}$$

Rearranging

$$\frac{(e^{ikh} - e^{-ikh})}{2h} - ik = \frac{(ik)^3 h^2}{6} e^{i\theta k}$$

Therefore defining

$$\psi_h^{\mathcal{F}} = \frac{S_h - S_{-h}}{2h}, \quad (\psi_h^{\mathcal{F}} \hat{f})_k = \frac{(e^{ikh} - e^{-ikh})}{2h} \hat{f}_k = \frac{i \sin(kh)}{h} \hat{f}_k$$

And then

$$\|(\Phi_h^{\mathcal{F}} - D)\hat{f}\|_{\ell^2} = \left(\sum_{k \in \mathbb{Z}} \left| \frac{(ik)^3 h^2}{6} e^{i\theta k} \right|^2 |\hat{f}_k|^2 \right)^{\frac{1}{2}} = \frac{h^2}{6} \|D^2 \hat{f}\|_{\ell^2}$$

But then the question poses itself, why don't we immediately just use D and must discretize?

Reason 1 — You Don't Have $\hat{f}_k$ for All k

In practice, you never observe a function f through its exact Fourier coefficients. You observe it through a **finite set of point samples** $\{f(x_0), f(x_1), \dots, f(x_J)\}$ on a mesh. The map from point samples to Fourier coefficients requires solving a system — and for non-periodic or non-smooth f , the Fourier series may converge slowly or not at all (Gibbs phenomenon). The finite-difference operator $\Phi_h^{\mathcal{F}}$ works **directly on point values** without ever needing the full spectral decomposition.

Reason 2 — D is Unbounded on $\mathcal{B}$

This is the deepest reason. D is **not a bounded operator** on $\ell^2(\mathbb{Z})$:

$$\|D\hat{f}\|_{\ell^2}^2 = \sum_k k^2 |\hat{f}_k|^2,$$

which is **infinite** for a generic $\hat{f} \in \ell^2$. You need $\hat{f} \in \mathcal{B}_1$ just for $D\hat{f}$ to exist.

Meanwhile $\Phi_h^{\mathcal{F}}$ is bounded on all ℓ^2 with norm 2. So D is a **densely defined, unbounded, closed operator** — a completely different beast from a numerical method's perspective. You can't implement an unbounded operator on a computer.

Reason 3 — The PDE Lives in Physical Space

The heat equation $\partial_t u - \kappa(x) \partial_{xx} u = f$ has a **variable coefficient** $\kappa(x)$. In physical space this is a simple pointwise multiplication. In Fourier space it becomes a **convolution** $\hat{\kappa} * \hat{u}$ — coupling every mode to every other mode, turning a sparse local stencil into

a **dense global operator**. The finite-difference discretization preserves sparsity; the spectral approach destroys it unless κ is constant.

Reason 4 — Computational Cost

Even when κ is constant and D is diagonal in Fourier space, computing $\hat{f}_k$ from point samples costs $O(N \log N)$ via the FFT. For a single time, step in a PDE solver, applying a finite-difference stencil costs only $O(N)$ directly on the mesh — no transform needed.

Reason 5 — Boundary Conditions Break Fourier Diagonalization

The heat equation on $[0, 1]$ with Dirichlet conditions $u(0) = u(1) = 0$ is **not periodic** — the Fourier basis e^{ikx} doesn't diagonalize ∂_{xx} on this domain. You'd need sine series, and even then the variable-coefficient case resists spectral diagonalization entirely.

Part II — Heat Equation in Fourier Space with Complex Conductivity

Let us extend the heat equation to the complex-valued conductivity. Consider the PDE on $\mathbb{T} \times [0, T]$.

$$\partial_t u - \kappa(x) \partial_{xx} u = f(t, x), \quad u(x, 0) = u_0(x),$$

Where $\kappa \in L^\infty(\mathbb{T}, \mathbb{C})$ satisfies $\operatorname{Re}(\kappa(x)) \geq \kappa_0 > 0$ almost everywhere. Working in Fourier space with $u(t, \cdot) = \sum_k \hat{u}_k(t) e^{ikx}$ & $\kappa(x) = \sum_k \hat{\kappa}_j e^{ikx}$, the PDE becomes convolution-operator ODE in ℓ^2 .

THE Θ -SCHEME IN ℓ^2

The Fourier-Space θ – *scheme* reads:

$$\frac{\hat{u}_k^{n+1} - \hat{u}_k^n}{\Delta k} + k^2 [(1 - \theta)(\hat{\kappa} * \hat{u}^n)_k + \theta(\hat{\kappa} * \hat{u}^{n+1})_k] = \hat{f}_k^{n+\frac{1}{2}}$$

If $\kappa(x) = \kappa_0 \in \mathbb{R}$ (constant), then:

$$\hat{\kappa}_j = \kappa_0 \cdot \delta_{j,0},$$

so the convolution collapses to:

$$(\hat{\kappa} * \hat{u})_k = \sum_j \kappa_0 \delta_{j,0} \hat{u}_{k-j} = \kappa_0 \hat{u}_k$$

The dense Toeplitz matrix becomes **diagonal**, modes decouple completely, and the θ -scheme reduces to independent scalar ODEs — one per mode k :

$$\frac{\hat{u}_k^{n+1} - \hat{u}_k^n}{\Delta t} + \kappa_0 k^2 [(1 - \theta)\hat{u}_k^n + \theta\hat{u}_k^{n+1}] = \hat{f}_k^{n+\frac{1}{2}}.$$

Part (a) — CFL Condition from Spectral Stability

When κ_0 is constant and real, the scheme decouples mode-by-mode. For the explicit scheme

$$\hat{u}_k^{n+1} = \hat{u}_k^n - \kappa_0 k^2 \Delta t \hat{u}_k^n + \hat{f}_k^{n+\frac{1}{2}}, \quad \hat{u}_k^{n+1} = (1 - \kappa_0 k^2 \Delta t) \hat{u}_k^n + \hat{f}_k^{n+\frac{1}{2}},$$

And therefore, we're dealing with a system of the form

$$\hat{U}^{n+1} = A \hat{U}^n + F^n$$

With A a diagonal matrix, now for stability we require of course that

$$\rho(A) < 1 \Leftrightarrow |1 - \kappa_0 k^2 \Delta t| \leq 1$$

Since $\kappa_0 > 0$, the factor $1 - \kappa_0 k^2 \Delta t$ is **real**, so:

$$-1 \leq 1 - \kappa_0 k^2 \Delta t \leq 1.$$

The right inequality $1 - \kappa_0 k^2 \Delta t \leq 1$ is always satisfied (since $\kappa_0, k^2, \Delta t \geq 0$).

The **binding constraint** is the left one:

$$1 - \kappa_0 k^2 \Delta t \geq -1 \Rightarrow \kappa_0 k^2 \Delta t \leq 2.$$

This must hold for **all** $k \in \mathbb{Z}$, so it is controlled by $k_{\max}$.

On a mesh of spacing $\Delta x = 1/(J + 1)$, by the Nyquist–Shannon principle the highest representable mode is:

$$k_{\max} = \frac{\pi}{\Delta x},$$

so $k_{\max} = \frac{\pi^2}{(\Delta x)^2}$. The stability condition becomes:

$$\kappa_0 \frac{\pi^2}{(\Delta x)^2} \Delta t \leq 2 \Rightarrow \kappa_0 \frac{\Delta t}{(\Delta x)^2} \leq \frac{2}{\pi^2}.$$

Dropping the π^2 factor (it appears because of how you normalize k ; in the course's finite-difference indexing where the highest mode contributes -2 to the second difference, the constant comes out as $1/2$), you recover:

$$\boxed{\kappa_0 \frac{\Delta t}{(\Delta x)^2} \leq \frac{1}{2}},$$

Part (b) Implicit Scheme with Complex Conductivity

Let κ be complex-valued (we impose $\text{Re}(\kappa) \geq \kappa_0 > 0$). The multiplication operator $M_\kappa: L^2(\mathbb{T}) \rightarrow L^2(\mathbb{T})$, $(M_\kappa u)(x) = \kappa(x)u(x)$, is bounded but not self-adjoint.

The form of is:

$$a(u, v) = \int_{\mathbb{T}} \kappa(x) \partial_x u \partial_x^- v dx.$$

This takes two functions $u, v \in H^1(\mathbb{T})$ and returns a **complex number**. It is called **sesquilinear** because it is:

- Linear in the first argument u
- **Conjugate-linear** in the second argument v (due to the $\partial_x^- v$)

Think of it as a "twisted" inner product, weighted by $\kappa(x)$.

is the **Sobolev space** of functions on the torus $\mathbb{T}$ that are in L^2 and whose first derivative is also in L^2 :

$$H^1(\mathbb{T}) = \{u \in L^2(\mathbb{T}): \partial_x u \in L^2(\mathbb{T})\},$$

with inner product and norm:

$$\langle u, v \rangle_{H^1} = \langle u, v \rangle_{L^2} + \langle \partial_x u, \partial_x v \rangle_{L^2}, \quad \|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\partial_x u\|_{L^2}^2.$$

In Fourier space, if $u = \sum_k \hat{u}_k e^{ikx}$, then:

$$\|u\|_{H^1}^2 = \sum_{k \in \mathbb{Z}} (1 + k^2) |\hat{u}_k|^2.$$

So an H^1 function is a signal whose **size** and whose **slope** are both square-integrable. Intuitively:

- If $\partial_x u$ is in L^2 , the function cannot oscillate too violently or have too many sharp jumps.
- The norm

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\partial_x u\|_{L^2}^2$$

measures “energy of the signal + energy of its slope”.

So H^1 is a **Hilbert space of moderately smooth signals**: more regular than arbitrary L^2 functions, but far less restrictive than C^1 .

Fourier-space picture

On the torus $\mathbb{T}$, every u has a Fourier series

$$u(x) = \sum_{k \in \mathbb{Z}} \hat{u}_k e^{ikx}.$$

- Being in L^2 means

$$\sum_{k \in \mathbb{Z}} |\hat{u}_k|^2 < \infty.$$

- Differentiating multiplies the k -th mode by ik , so

$$\partial_x u(x) = \sum_{k \in \mathbb{Z}} (ik) \hat{u}_k e^{ikx}.$$

Then

$$\|\partial_x u\|_{L^2}^2 = \sum_{k \in \mathbb{Z}} k^2 |\hat{u}_k|^2. \quad \square$$

Putting this into the H^1 -norm gives

$$\|u\|_{H^1}^2 = \sum_{k \in \mathbb{Z}} (1 + k^2) |\hat{u}_k|^2.$$

So:

- High frequencies (large $|k|$) get penalized by a factor $1 + k^2$.
- For the sum to converge, $|\hat{u}_k|$ must decay faster than in plain L^2 .

The implicit scheme ($\theta = 1$) in physical space says:

$$\frac{u^{n+1} - u^n}{\Delta t} - \kappa(x) \partial_{xx} u^{n+1} = f^{n+1/2}.$$

This is called the **strong form** — it requires u^{n+1} to satisfy the equation **pointwise**, at every $x \in \mathbb{T}$. This demands a lot of regularity (u^{n+1} must be twice differentiable everywhere).

Take any $v \in H^1(\mathbb{T})$ and **multiply both sides** by $v(\bar{x})$:

$$\frac{u^{n+1} - u^n}{\Delta t} \bar{v} - \kappa(x) \partial_{xx} u^{n+1} \bar{v} = f^{n+1/2} \bar{v}.$$

Integrate both sides over $\mathbb{T}$:

$$\int_{\mathbb{T}} \frac{u^{n+1} - u^n}{\Delta t} \bar{v} dx - \int_{\mathbb{T}} \kappa(x) \partial_{xx} u^{n+1} \bar{v} dx = \int_{\mathbb{T}} f^{n+1/2} \bar{v} dx.$$

The first and last integrals are already L^2 inner products:

$$\frac{1}{\Delta t} \langle u^{n+1} - u^n, v \rangle_{L^2} - \int_{\mathbb{T}} \kappa(x) \partial_{xx} u^{n+1} \bar{v} dx = \langle f^{n+1/2}, v \rangle_{L^2}.$$

Apply **integration by parts** (on $\mathbb{T}$, the boundary terms vanish because everything is periodic):

$$\begin{aligned} \int_{\mathbb{T}} \kappa(x) \partial_{xx} u^{n+1} \bar{v} dx &= [\kappa(x) \partial_x u^{n+1} \bar{v}]_{\partial \mathbb{T}} - \int_{\mathbb{T}} \kappa(x) \partial_x u^{n+1} \partial_x \bar{v} dx. \\ &= 0 \text{ (periodicity)} \end{aligned}$$

So:

$$- \int_{\mathbb{T}} \kappa(x) \partial_{xx} u^{n+1} \bar{v} dx = + \int_{\mathbb{T}} \kappa(x) \partial_x u^{n+1} \partial_x \bar{v} dx = a(u^{n+1}, v).$$

Substituting back:

$$\frac{1}{\Delta t} \langle u^{n+1} - u^n, v \rangle_{L^2} + a(u^{n+1}, v) = \langle f^{n+1/2}, v \rangle_{L^2}.$$

Multiply through by Δt :

$$\langle u^{n+1}, v \rangle_{L^2} - \langle u^n, v \rangle_{L^2} + \Delta t a(u^{n+1}, v) = \Delta t \langle f^{n+1/2}, v \rangle_{L^2}.$$

Rearranging:

$$\boxed{\langle u^{n+1}, v \rangle_{L^2} + \Delta t a(u^{n+1}, v) = \langle u^n + \Delta t f^{n+1/2}, v \rangle_{L^2}.}$$

Theorem (Lax–Milgram). Let V be a Hilbert space and $a: V \times V \rightarrow \mathbb{C}$ a sesquilinear form. Suppose:

1. **Continuity:** $|a(u, v)| \leq C \|u\|_V \|v\|_V$ for some constant $C > 0$.

2. **Coercivity:** $\operatorname{Re}(a(u, u)) \geq \alpha \|u\|_V^2$ for some constant $\alpha > 0$.

Then for every continuous linear functional $F: V \rightarrow \mathbb{C}$, there exists a **unique** $u \in V$ such that $a(u, v) = F(v)$ for all $v \in V$.

Think of it as the abstract generalization of: "an invertible matrix A always gives a unique solution to $Ax = b$." Here the "matrix" is $a(\cdot, \cdot)$ and the "vector b " is $F(v)$.

Then the problem we're trying to solve becomes

$$\boxed{\langle u^{n+1}, v \rangle_{L^2} + \Delta t a(u^{n+1}, v) = \langle u^n + \Delta t f^{n+1/2}, v \rangle_{L^2}.}$$

The left side defines a **new sesquilinear form**: $b(u, v) = \langle u, v \rangle_{L^2} + \Delta t \cdot a(u, v)$,

1. b is bounded:

$$|b(u, v)| = |\langle u, v \rangle_{L^2} + \Delta t \cdot a(u, v)| \leq |\langle u, v \rangle_{L^2}| + \Delta t |a(u, v)| \leq \|u\| \|v\| + \Delta t |a(u, v)|$$

But:

$$\begin{aligned} |a(u, v)| &= \left| \int_{\mathbb{T}} \kappa(x) \partial_x u \partial_x^- v \, dx \right| \leq \int_{\mathbb{T}} |\kappa(x)| |\partial_x u| |\partial_x^- v| \, dx \\ &\leq \|\kappa\|_{\infty} \int_{\mathbb{T}} |\partial_x u| |\partial_x^- v| \, dx \leq \|\kappa\|_{\infty} \|\partial_x u\| \|\partial_x v\| \end{aligned}$$

Therefore,

$$\begin{aligned} |b(u, v)| &\leq \|u\| \|v\| + \Delta t \|\kappa\|_{\infty} \|\partial_x u\| \|\partial_x v\| \\ &\leq \|u\|_{H_1} \|v\|_{H_1} + \Delta t \|\kappa\|_{\infty} \|u\|_{H_1} \|v\|_{H_1} \\ &= (1 + \Delta t \|\kappa\|_{\infty}) \|u\|_{H_1} \|v\|_{H_1} \end{aligned}$$

And

$$|b(u, v)| \leq (1 + \Delta t \|\kappa\|_{\infty}) \|u\|_{H_1} \|v\|_{H_1}$$

2. **Coercivity:** $\operatorname{Re}(b(u, u)) \geq \alpha \|u\|_{H_1}^2$

$$b(u, u) = \langle u, u \rangle_{L^2} + \Delta t \cdot a(u, u), = \|u\|^2 + \Delta t \int_{\mathbb{T}} \kappa(x) |\partial_x u|^2 \, dx$$

$$\operatorname{Re}(b(u, u)) = \|u\|^2 + \Delta t \operatorname{Re} \left(\int_{\mathbb{T}} \kappa(x) |\partial_x u|^2 \, dx \right)$$

But,

$$\operatorname{Re} \left(\int_{\mathbb{T}} \kappa(x) |\partial_x u|^2 \, dx \right) = \int_{\mathbb{T}} \operatorname{Re}(\kappa(x)) |\partial_x u|^2 \, dx$$

because $|\partial_x u|^2 \geq 0$ is **real**, so $\operatorname{Re}(\kappa \cdot |\partial_x u|^2) = \operatorname{Re}(\kappa) \cdot |\partial_x u|^2$.

$$\operatorname{Re}(b(u, u)) = \|u\|^2 + \Delta t \int_{\mathbb{T}} \operatorname{Re}(\kappa(x)) |\partial_x u|^2 \, dx \geq \|u\|^2 + \Delta t \kappa_0 \|\partial_x u\|^2$$

Letting $\tau = \min\{1, \Delta t \kappa_0\}$

$$\operatorname{Re}(b(u, u)) \geq \tau(\|u\|^2 + \|\partial_x u\|^2) = \tau \|u\|_{H_1}^2$$

I think by now you'd agree on how useful this norm is.

Regardless now, by **Lax–Milgram**, the problem

$$\boxed{\langle u^{n+1}, v \rangle_{L^2} + \Delta t \, a(u^{n+1}, v) = \langle u^n + \Delta t f^{n+1/2}, v \rangle_{L^2}.}$$

Now admits a unique solution $\forall v$, as $F(v) = \langle \underbrace{u^n + \Delta t f^{n+1/2}}_{\text{fixed and predetermined}}, v \rangle_{L^2}$.

So that our Implicit scheme is well-posed and solvable.

PART III - Geodesic Finite Differences on a Riemannian Manifold

Let $(\mathcal{M}, g)$ be a complete Riemannian manifold of $\dim M = d$, *Levi – Civita* connection ∇ , Riemannian exponential map $\exp_p: T_p \mathcal{M} \rightarrow \mathcal{M}$, the “step size” h is now a geodesic arclength.

Suppose $f \in C^3(\mathcal{M})$, On a Riemannian manifold $(\mathcal{M}, g)$, a geodesic is a curve $\gamma: I \rightarrow \mathcal{M}$ that satisfies:

$$\frac{D\dot{\gamma}}{dt} = 0,$$

where $\frac{D}{dt}$ is the **covariant derivative along γ** . In local coordinates $(x^1, \dots, x^d)$, this reads:

$$\ddot{\gamma}^k + \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0, k = 1, \dots, d,$$

where Γ_{ij}^k are the **Christoffel symbols** of the Levi-Civita connection, determined entirely by the metric g :

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}).$$

This is a **second-order ODE system** in d unknowns. By the **Picard–Lindelöf theorem**, given initial conditions:

$$\gamma(0) = p \in \mathcal{M}, \dot{\gamma}(0) = v \in T_p \mathcal{M},$$

there exists a **unique** geodesic $\gamma_{p,v}: (-\epsilon, \epsilon) \rightarrow \mathcal{M}$ for some $\epsilon > 0$.

What is the exponential map?

at each point $p \in \mathcal{M}$, we have a tangent space: $T_p \mathcal{M}$

Let $v \in T_p \mathcal{M}$, we have a unique Geodesic satisfying: $\gamma_v(0) = p, \gamma'_v(0) = v$

The exponential $\exp_p(v) = \gamma_v(1)$, becomes a well-defined map:

$$\begin{aligned} \exp_p : T_p \mathcal{M} &\rightarrow \mathcal{M} \\ v &\rightarrow \gamma_v(1) \end{aligned}$$

Examples

Let $\mathcal{M}$ be a flat plane inside $\mathbb{R}^3$

Geodesics are still straight lines inside the plane

So if $p \in \mathcal{M}$, and $v \in T_p \mathcal{M}$ then

$$\exp_p(v) = p + v$$

But consider now $\mathcal{S}^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$.

At a point $p \in \mathcal{S}^2$, the tangent space is the plane perpendicular to p :

$$T_p \mathcal{S}^2 = \{v \in \mathbb{R}^3 : v \cdot p = 0\}$$

If you start at p with tangent velocity v , then our geodesic will be

$$\gamma(t) = \cos(rt) p + \sin(rt) \frac{v}{r}$$

So

$$\exp_p(v) = \cos(r) p + \sin(r) \frac{v}{r}$$

So given a geodesic γ ,

$$p = \gamma(0), \quad v = \dot{\gamma}(0)$$

So $\exp_{\gamma(t)}(h\dot{\gamma}(0))$ corresponds to the step size of h in the direction of v .

Letting $c(t) = \exp_{\gamma(t_0)}(tv) = \gamma(t)$

$$F(t) = f \circ c(t) \Rightarrow F'(t) = df(c(t)) \circ c'(t)$$

$$\text{So } \exp_{\gamma(t)}(h\dot{\gamma}(0)) = f(p) + h df_p(v) + \frac{h^2}{2} \text{Hess}_p f(p)(v, v) + O(h^3)$$

Now finite difference methods dealing with manifold deal with this finite step of

$$\Delta_h^g f = \frac{1}{h} \left(f(\exp_{\gamma(t)}(h\dot{\gamma}(t))) - f(\gamma(t)) \right)$$

$$\frac{1}{h} [\exp_{\gamma(t)}(h\dot{\gamma}(t)) - f(\gamma(t))] = df_p(v) + \frac{h}{2} \text{Hess}_p f(p)(v, v) + O(h^2)$$

So we're approximating the derivative at a specific direction,

A differential equation on a manifold usually looks like

$$\dot{x}(t) = F(x(t)), \quad x(t) \in \mathcal{M}$$

Where F is a vector field on the manifold.

For example, the motion of a particle on a manifold, then given the nature of the motion and some coordinates we can construct a differential equation of this sorts to attempt and solve it numerically.

In flat space, we use Euler: In $\mathbb{R}^n$, for

$$\dot{x}(t) = F(x(t)),$$

Euler's Method is

$$x_{n+1} = x_n + hF(x_n)$$

But on a manifold

$$x_n \in \mathcal{M}, \quad F(x_n) \in T_{x_n} \mathcal{M}.$$

A point and a tangent vector are not objects you can simply add, then we rely on exponentials for the notion of a step of length h towards $F(x_n)$

$$x_{n+1} = \exp_{x_n}(h(F(x_n)))$$

Let $\mathcal{M} = S^2$

Suppose

$$\dot{p}(t) = F(p(t)), \quad p(t) \in S^2$$

At every point p , the vector $F(p)$ must be tangent to the sphere:

$$F(p) \cdot p = 0$$

The manifold step is:

$$p_{n+1} = \exp_{p_n}(hF(p_n))$$

For the unit sphere, we know So, $p_{n+1} = \exp_p(v) = \cos(h\|F(p_n)\|)p + \sin(h\|F(p_n)\|)\frac{F(p_n)}{\|F(p_n)\|}$

The **weak/variational formulation** is the same idea as in Euclidean FEM, but with Sobolev spaces built from g : find $u \in H^1(M)$ such that

$$a(u, v) = \ell(v) \forall v \in H^1(M),$$

with

$$a(u, v) = \int_M \langle A \nabla u, \nabla v \rangle_g + cuv \, d\text{vol}_g, \ell(v) = \int_M f v \, d\text{vol}_g.$$

So the *analytic* theory (Lax–Milgram, elliptic regularity, spectral properties of Δ_g , etc.) is essentially the same as in $\mathbb{R}^n$, but with:

- metric g entering gradients, divergence, and volume form
- manifold topology affecting boundary conditions, eigenstructure, etc

Let's take a look at the strong form again,

$$-\nabla \cdot (A \nabla u) + cu = f \quad \text{on } M,$$

- ∇u is the Riemannian gradient (a vector field) defined using g .
- $\nabla \cdot (\cdot)$ is the Riemannian divergence (trace of covariant derivative, or via Lie derivative of the volume form).
- A is a symmetric, positive definite (1,1)-tensor field (think anisotropic diffusion tensor).
- $c \geq 0$ is a scalar function.
- f is a given source function.

In local coordinates $(x^1, \dots, x^n)$, this looks like

$$-\frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} A^{ij} \partial_j u) + cu = f,$$

with the usual summation convention and $g = \det(g_{ij})$.

This is the “strong form”: it makes sense pointwise if u is twice differentiable.

2. Sobolev space $H^1(M)$

On a compact Riemannian manifold you can define Sobolev spaces exactly as in $\mathbb{R}^n$: patch with charts, use a partition of unity, and glue Euclidean Sobolev norms.

For $H^1(M)$:

- Start with $C^\infty(M)$.
- In local coordinates, derivatives of order ≤ 1 are square-integrable.
- The norm is equivalent to

$$\|u\|_{H^1(M)}^2 = \int_M |u|^2 d\text{vol}_g + \int_M \|\nabla u\|_g^2 d\text{vol}_g.$$

Then $H^1(M)$ is the completion of $C^\infty(M)$ under this norm.

Assume for the moment u is smooth and solves the strong PDE. Take any smooth test function $v \in C^\infty(M)$. Multiply the PDE by v and integrate over M :

$$\int_M (-\nabla \cdot (A\nabla u) + cu) v d\text{vol}_g = \int_M f v d\text{vol}_g.$$

Split the integral:

$$-\int_M (\nabla \cdot (A\nabla u)) v d\text{vol}_g + \int_M cuv d\text{vol}_g = \int_M f v d\text{vol}_g.$$

Now use the **Riemannian integration by parts** identity (divergence theorem): for any vector field X ,

$$\int_M (\nabla \cdot X) v d\text{vol}_g = - \int_M \langle X, \nabla v \rangle_g d\text{vol}_g,$$

provided M has no boundary or v vanishes on the boundary.

provided M has no boundary or v vanishes on the boundary.

Apply with $X = A\nabla u$:

$$-\int_M (\nabla \cdot (A\nabla u)) v d\text{vol}_g = \int_M \langle A\nabla u, \nabla v \rangle_g d\text{vol}_g.$$

So the equation becomes

$$\int_M \langle A\nabla u, \nabla v \rangle_g d\text{vol}_g + \int_M cuv d\text{vol}_g = \int_M f v d\text{vol}_g$$

This is the weak form in informal notation. Now package it:

- Define the bilinear form

$$a(u, v) := \int_M \langle A\nabla u, \nabla v \rangle_g + cuv d\text{vol}_g.$$

- Define the linear functional

$$\ell(v) := \int_M f v \, d\text{vol}_g.$$

Then the weak formulation is:

Find $u \in H^1(M)$ such that

$$a(u, v) = \ell(v) \forall v \in H^1(M).$$

Here, u and v are only required to be in H^1 , so “ ∇u ” is interpreted as a **weak gradient** (in distributional sense), and integration by parts is built into the definition of that weak gradient.

To use this formulation (for theory or numerics), you want existence and uniqueness of $u \in H^1(M)$. This is where **Lax–Milgram** enters.

Lax–Milgram (Hilbert-space version):

Let V be a Hilbert space (here $V = H^1(M)$). Let $a: V \times V \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be bilinear (sesquilinear), **continuous** and **coercive**:

- Continuity: $|a(u, v)| \leq C \|u\|_V \|v\|_V$.
- Coercivity: $a(u, u) \geq \alpha \|u\|_V^2$ for some $\alpha > 0$.

Let $\ell: V \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be continuous linear functional. Then there exists a unique $u \in V$ such that

$$a(u, v) = \ell(v) \quad \forall v \in V.$$

For numerical analysis on manifolds, the weak form is central because:

- Only **first derivatives** appear (through ∇u), which is good for finite elements, especially on curved geometries where second derivatives are ugly.
- Geometry appears naturally in g and $d\text{vol}_g$, and this is exactly what you approximate by a mesh/NURBS/point cloud.
- The bilinear form $a(u, v)$ is what you discretize to get the stiffness matrix (e.g. surface FEM for Laplace–Beltrami).

For example, for the Laplace–Beltrami equation $-\Delta_g u = f$ on a closed surface $\Gamma \subset \mathbb{R}^3$, the weak form is

$$\int_{\Gamma} \langle \nabla_{\Gamma} u, \nabla_{\Gamma} v \rangle dS = \int_{\Gamma} f v dS,$$

In your quantum context, exactly the same idea will apply:

- The Hamiltonian $-\frac{\hbar^2}{2m}\Delta_g + V$ gives a bilinear form

$$a(\psi, \phi) = \frac{\hbar^2}{2m} \int_M \langle \nabla \psi, \nabla \phi \rangle_g d\text{vol}_g + \int_M V \psi \phi d\text{vol}_g,$$

and this is what you approximate when building a discrete Hamiltonian that's self-adjoint in the discrete L^2 inner product defined by the metric.

As a learning example, we model a spinless quantum particle of mass $m > 0$

For quantum mechanics on S^2 , the natural Hilbert space is

$$L^2(S^2) = \left\{ \psi: S^2 \rightarrow \mathbb{C} \mid \int_{S^2} |\psi|^2 d\text{vol}_g < \infty \right\}$$

with **complex** inner product

$$\langle \psi, \phi \rangle_{L^2} = \int_{S^2} \bar{\psi} \phi d\text{vol}_g.$$

Similarly, the energy space is the complex Sobolev space

$$H^1(S^2; \mathbb{C}) = \{ \psi \in L^2(S^2; \mathbb{C}) : \nabla \psi \in L^2(TS^2 \otimes \mathbb{C}) \},$$

with norm

$$\| \psi \|_{H^1}^2 = \int_{S^2} |\psi|^2 d\text{vol}_g + \int_{S^2} \| \nabla \psi \|_g^2 d\text{vol}_g.$$

So the **space** where the weak formulation lives is now a complex Hilbert space $V = H^1(S^2; \mathbb{C})$. The *definition of H^1* is the same as in the real case, just allowing complex values.

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So the **space** where the weak formulation lives is now a complex Hilbert space $V = H^1(S^2; \mathbb{C})$. The *definition of H^1* is the same as in the real case, just allowing complex values.

For a *real-valued* function $u: S^2 \rightarrow \mathbb{R}$:

- At each point p , $\nabla u(p) \in T_p S^2$, a real 2-dimensional vector space.
- The collection ∇u is a **real** vector field, an element of $L^2(TS^2)$ if it is square-integrable in norm.

For a *complex-valued* function $\psi: S^2 \rightarrow \mathbb{C}$:

- You can write $\psi = u + iv$ with u, v real-valued.
- Define $\nabla \psi := \nabla u + i \nabla v$.

Now $\nabla u(p)$ and $\nabla v(p)$ are both in the real tangent space $T_p S^2$. The combination

$$\nabla \psi(p) = \nabla u(p) + i \nabla v(p)$$

is a **complex linear combination of real tangent vectors**. So at each point p , $\nabla\psi(p)$ naturally lives in

$$T_p S^2 \otimes_{\mathbb{R}} \mathbb{C},$$

the **complexified tangent space**.

That's all $TS^2 \otimes \mathbb{C}$ means: “take the real tangent bundle and allow complex coefficients.”

2. Why write $L^2(TS^2 \otimes \mathbb{C})$?

The Sobolev space $H^1(S^2; \mathbb{C})$ is defined as

$$H^1(S^2; \mathbb{C}) = \{ \psi \in L^2(S^2; \mathbb{C}) : \nabla\psi \in L^2(TS^2 \otimes \mathbb{C}) \},$$

where:

- $L^2(S^2; \mathbb{C})$ means square-integrable complex-valued functions on S^2 ;
- $L^2(TS^2 \otimes \mathbb{C})$ means square-integrable complex-valued **vector fields** tangent to S^2 , i.e.

$$\int_{S^2} \| \nabla\psi(p) \|_g^2 \, d\text{vol}_g(p) < \infty.$$

Writing $TS^2 \otimes \mathbb{C}$ is just a clean way of saying “vector fields with complex coefficients but still tangent to the sphere.” All the geometry (metric, Levi–Civita connection, Laplace–Beltrami) is defined on the **real** tangent bundle; complexification is purely algebraic and doesn't change any formula.

PART I

Our metric on S^2 is given by the first fundamental form,

$$\begin{aligned} \sigma(\theta, \phi) &= (\cos(\theta) \sin(\phi), \sin(\theta) \sin(\phi), \cos(\phi)) \\ I &= \begin{bmatrix} \|\sigma_\theta\|^2 & 0 \\ 0 & \|\sigma_\phi\|^2 \end{bmatrix} = \begin{bmatrix} \sin^2(\phi) & 0 \\ 0 & 1 \end{bmatrix} = \sin^2(\phi) \, d\theta^2 + d\phi^2 \end{aligned}$$

As for now we derive the coordinate expression on the regular chart away from the poles, and handle the global sphere later by additional charts or pole regularity conditions, because the PDE itself is global on S^2 ; it is the coordinate system that breaks down at the poles

Globally, $H^1(S^2)$ is obtained by covering S^2 with regular charts and gluing the local H^1 descriptions by a smooth partition of unity.

So our volume form in this region will be

$$dvol_g = \sqrt{|g|} d\theta d\phi = \sin(\phi) d\theta d\phi$$

Given that

$$\|u\|_{H^1(S^2)}^2 = \int_{S^2} |u|^2 dvol_g + \int_{S^2} \|\nabla u\|_g^2 dvol_g.$$

We have

$$\begin{aligned} \nabla_g u &= [g^{\theta\theta} \partial_\theta u] \cdot \partial_\theta + [g^{\phi\phi} \partial_\phi u] \cdot \partial_\phi \\ &= \left[\frac{1}{\sin^2(\phi)} \partial_\theta u \right] \cdot \partial_\theta + [\partial_\phi u] \cdot \partial_\phi \end{aligned}$$

Then we can conclude

$$\|\nabla u\|_g^2 = \frac{1}{\sin^2(\phi)} (\partial_\theta u)^2 + (\partial_\phi u)^2$$

Therefore

$$\begin{aligned} \|u\|_{H^1(S^2)}^2 &= \int_{S^2} |u|^2 dvol_g + \int_{S^2} \|\nabla u\|_g^2 dvol_g \\ &= \int_{S^2} |u|^2 \sin(\phi) d\theta d\phi + \int_{S^2} \left[\frac{1}{\sin^2(\phi)} (\partial_\theta u)^2 + (\partial_\phi u)^2 \right] \sin(\phi) d\theta d\phi \end{aligned}$$

The Laplace–Beltrami operator on a 2D Riemannian manifold with metric (g_{ij}) is

$$\Delta_g u = \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} g^{ij} \partial_j u) = \frac{1}{\sin(\theta)} \partial_\theta (\sin(\theta) \partial_\theta u) + \frac{1}{\sin^2(\theta)} \partial_\phi \partial_\phi u.$$

Suppose ψ is the wave function of the particle moving within the sphere, then the wave equation is given by

$$-\frac{\hbar}{2m}\Delta_g\psi + V\psi = f$$

Multiplying both sides by a smooth test function $v \in C^\infty(S^2)$ and integrating over S^2 with respect to $d\text{vol}_g$, we get the resulting identity:

$$\int_{S^2} \left(-\frac{\hbar^2}{2m}\Delta_g\psi \right) v d\text{vol}_g + \int_{S^2} V\psi v d\text{vol}_g = \int_{S^2} f v d\text{vol}_g. \quad (*)$$

the Riemannian divergence theorem gives (which holds since S^2 has no boundary)

$$\int_{S^2} (\Delta_g\psi) v d\text{vol}_g = - \int_{S^2} \langle \nabla\psi, \nabla v \rangle_g d\text{vol}_g,$$

Combine this with the potential and source terms to obtain

$$\frac{\hbar^2}{2m} \int_{S^2} \langle \nabla\psi, \nabla v \rangle_g d\text{vol}_g + \int_{S^2} V\psi v d\text{vol}_g = \int_{S^2} f v d\text{vol}_g$$

Define the sesquilinear form $a(\cdot, \cdot)$ and linear functional $\ell(\cdot)$ on $H^1(S^2)$ by

$$a(u, v) := \frac{\hbar^2}{2m} \int_{S^2} \langle \nabla u, \nabla v \rangle_g d\text{vol}_g + \int_{S^2} Vuv d\text{vol}_g,$$

$$\ell(v) := \int_{S^2} f v d\text{vol}_g.$$

Then by Lax – Milgram, our goal is to find $\psi \in H^1(S^2)$ such that

$$a(\psi, v) = \ell(v) \forall v \in H^1(S^2).$$

- Is a bounded?

Use Cauchy–Schwarz in $L^2(S^2)$:

$$\left| \frac{\hbar^2}{2m} \int_{S^2} \langle \nabla u, \nabla v \rangle_g d\text{vol}_g \right| \leq \frac{\hbar^2}{2m} \|\nabla u\|_{L^2} \|\nabla v\|_{L^2}$$

Here $\|\nabla u\|_{L^2}^2 = \int_{S^2} \|\nabla u\|_g^2 d\text{vol}_g$.

$$|a| \leq \left| \frac{\hbar^2}{2m} \int_{S^2} \langle \nabla u, \nabla v \rangle_g d\text{vol}_g + \int_{S^2} Vuv d\text{vol}_g \right| \leq$$

$$\begin{aligned}
& \frac{\hbar^2}{2m} \left| \int_{S^2} \langle \nabla u, \nabla v \rangle_g \, d\text{vol}_g \right| + \int_{S^2} |Vuv| \, d\text{vol}_g \\
& \leq \frac{\hbar^2}{2m} \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + \|V\|_\infty \langle |u|, |v| \rangle_{L^2} \leq \frac{\hbar^2}{2m} \|\nabla v\|_2 \|\nabla u\|_2 + \|V\|_\infty \|u\|_2 \|v\|_2 \\
& \leq M(\|\nabla v\|_2 \|\nabla u\|_2 + \|u\|_2 \|v\|_2)
\end{aligned}$$

Then you can bound each factor by the full H^1 norm:

$$\|\nabla u\|_{L^2} \leq \|u\|_{H^1}, \quad \|u\|_{L^2} \leq \|u\|_{H^1}, \quad \text{and similarly, for } v.$$

So

$$|a(u, v)| \leq \left(\frac{\hbar^2}{2m} + \|V\|_{L^\infty} \right) \|u\|_{H^1(S^2)} \|v\|_{H^1(S^2)}.$$

- *Coersity?*

In our specific problem,

$$a(u, u) = \frac{\hbar^2}{2m} \int_{S^2} \|\nabla u\|_g^2 \, d\text{vol}_g + \int_{S^2} V |u|^2 \, d\text{vol}_g,$$

we rely on our assumption that and $V(x) \geq V_0 \geq 0$,

$$\begin{aligned}
\Re a(u, u) &= a(u, u) = \frac{\hbar^2}{2m} \int_{S^2} \|\nabla u\|_g^2 \, d\text{vol}_g + \int_{S^2} V |u|^2 \, d\text{vol}_g \\
&\geq \frac{\hbar^2}{2m} \int_{S^2} \|\nabla u\|_g^2 \, d\text{vol}_g + V_0 \int_{S^2} |u|^2 \, d\text{vol}_g \\
&\geq \alpha \left(\int_{S^2} \|\nabla u\|_g^2 \, d\text{vol}_g + \int_{S^2} |u|^2 \, d\text{vol}_g \right) = \alpha \|u\|_{H^1}^2
\end{aligned}$$

With $\alpha = \min\left(V_0, \frac{\hbar^2}{2m}\right)$

As we have

$$\ell(v) = \int_{S^2} f v \, d\text{vol}_g$$

one can clearly see it's linearity and boundedness, Then by Lax-Milgram, we have a unique solution to our posed problem (*)

LEARNING THE FINITE ELEMENT METHOD:

Consider

$$-u''(x) = f(x) \quad x \in]0,1[, \quad u(0) = u(1) = 0.$$

where:

$$- \quad u: \mathbb{R} \rightarrow \mathbb{R}, \quad f: \mathbb{R} \rightarrow \mathbb{R}$$

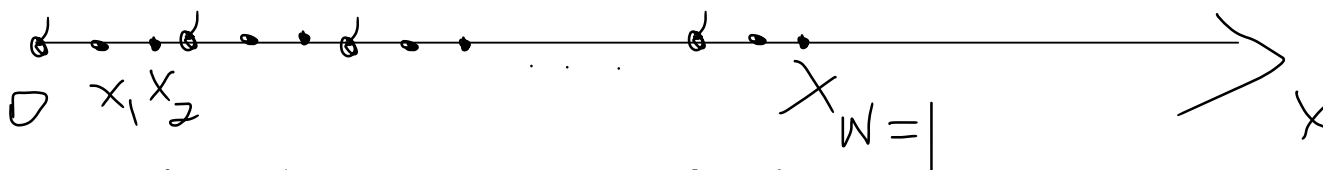
Our Sobolev space here is $H_0^1(]0,1[) = \{u \in L^2(]0,1[), \|u'(x)\|_2 < \infty\}$, of course with the additional zero boundary condition in the trace sense.

Multiply by a test function $v \in H_0^1(0,1)$, integrate over $(0,1)$, and integrate by parts to derive

$$\int_0^1 u'(x)v'(x) dx = \int_0^1 f(x)v(x) dx, \forall v \in H_0^1(0,1). \quad (**)$$

I believe we talked extensively about Lax Milgram, now our goal is to find our u satisfying (**).

Let's do something reminiscent of the finite difference method



Fix a uniform grid $0 = x_0 < x_1 < \dots < x_N = 1$, $h = 1/N$.

1. We now **hat functions** φ_i such that:

- $\varphi_i(x_j) = \delta_{ij}$
- each φ_i is piecewise linear on $[x_{i-1}, x_i] \cup [x_i, x_{i+1}]$.

A standard formula is:

$$\phi_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}}, & x \in [x_{i-1}, x_i], \\ \frac{x_{i+1} - x}{x_{i+1} - x_i}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases}$$

Define the finite element space

$$V_h = \text{span}\{\varphi_1, \dots, \varphi_{N-1}\} \subset H_0^1(0,1).$$

now what we would want to do here, is discretize the solution using our new set of basis elements. define

$$u_h = \sum_{j=1}^{n-1} U_j \phi_j$$

Find $u_h \in V_h$ such that

$$a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h$$

$$\int_0^1 u'(x) v'(x) dx = \int_0^1 f(x) v(x) dx, \quad \forall v_h \in V_h$$

$$\int_0^1 \sum_{j=1}^{n-1} U_j \phi_j'(x) \phi_i'(x) dx = \int_0^1 f(x) \phi_i(x) dx$$

$$\sum_{j=1}^{n-1} \left[\int_0^1 \phi_j'(x) \phi_i'(x) dx \right] U_j = \int_0^1 f(x) \phi_i(x) dx$$

$$\sum_{j=1}^{n-1} K_{ij} U_j = F_i$$

Where

$$K_{ij} = \int_0^1 \phi_j'(x) \phi_i'(x) dx, \quad F_i = \int_0^1 f(x) \phi_i(x) dx.$$

Which gives us a linear system, solvable using basic linear algebra.

Now, bars are easy to discretize not much of thought is given in the one dimensional case, but we're working with a sphere, and so how would discretizing a sphere look like?

To discretize S^2 , choose a finite set of vertices $x_i \in S^2$, connect them by triangles, and form a polyhedral surface

$$\Gamma_h = \bigcup_{T \in \mathcal{T}_h} T$$

made of flat planar triangles. In surface FEM, Γ_h is the geometric approximation of the smooth surface, and the triangle collection $\mathcal{T}_h$ is the mesh.

For the sphere, this is often done by starting from an icosahedron and recursively subdividing each triangle, then projecting the new vertices back onto the sphere, which improves mesh quality while keeping all vertices on S^2

In code, the mesh is usually stored as:

- A vertex array $X \in \mathbb{R}^{N \times 3}$, where row i is vertex x_i .
- A triangle array $K \in \mathbb{N}^{M \times 3}$, where row k contains the three vertex indices of triangle T_k .

$$V_h = \{ v_h \in C_0(\Gamma_h): v_h \mid T \text{ is affine linear on each triangle } T \in \mathcal{T}_h \}$$

These are globally continuous, piecewise linear functions on the triangulated surface. A nodal basis $\{\phi_1, \dots, \phi_N\}$ is built exactly like in 1D, except now each hat function is attached to a vertex: $\phi_i(x_j) = \delta_{ij}$, ϕ_i is linear on each incident triangle, and ϕ_i vanishes outside the patch of triangles touching vertex i .

So the discrete solution has the form

$$u_h = \sum_{j=1}^N U_j \phi_j.$$

The unknowns are the nodal coefficients U_j

Take one triangle T with vertices $x_a, x_b, x_c \in \mathbb{R}^3$. Since T is flat, everything on that element reduces to standard planar FEM on that triangle embedded in $\mathbb{R}^3$.

You compute:

- Edge vectors $e_1 = x_b - x_a$, $e_2 = x_c - x_a$.
- Triangle area

$$|T| = \frac{1}{2} \|e_1 \times e_2\|.$$

- A local affine map from the reference triangle $\hat{T}$ to T .

This is the key simplification: the manifold is curved globally, but each element is treated as a flat triangle locally.

Start from an icosahedron, because it is already a triangular approximation of the sphere, and refine it by subdividing each triangle and projecting the new vertices back to the unit sphere. A standard refinement takes a triangle with vertices $a, b, c \in S^2$, forms the edge midpoints, normalizes them back to radius 1, and replaces the old triangle by four new triangles.

Explicitly, define

$$m_{ab} = \frac{a+b}{\|a+b\|}, m_{bc} = \frac{b+c}{\|b+c\|}, m_{ca} = \frac{c+a}{\|c+a\|}.$$

Then replace (a, b, c) by the four triangles

$$(a, m_{ab}, m_{ca}), (m_{ab}, b, m_{bc}), (m_{ca}, m_{bc}, c), (m_{ab}, m_{bc}, m_{ca}).$$

This is exactly the usual icosphere subdivision

Once the triangulated sphere Γ_h is built, define

$$V_h = \{v_h \in C^0(\Gamma_h): v_h|_T \text{ is affine linear on every triangle } T\}.$$

Now let's go ahead and formulate the problem we're solving, Our continuous form on the true sphere is

$$a(u, v) = \frac{\hbar^2}{2m} \int_{S^2} \langle \nabla u, \nabla v \rangle_g d\text{vol}_g + \int_{S^2} V u \bar{v} d\text{vol}_g,$$

the gradient term should also be Hermitian, meaning

$$\langle \nabla u, \nabla v \rangle_g \text{ really behaves like } g(\nabla u, \overline{\nabla v}).$$

On the triangulated sphere, you no longer use the exact smooth metric directly. Instead, on each flat triangle $T \subset \Gamma_h$, the role of g is played by the **local flat metric** coming from the embedding of that triangle in $\mathbb{R}^3$, usually through the Jacobian J and Gram matrix $G = J^T J$.

So the exact metric is being replaced by a **piecewise-flat approximate metric**

$$a_h(u_h, v_h) = \frac{\hbar^2}{m^2} \int_{\Gamma_h} \nabla_{\Gamma_h} u_h \cdot \overline{\nabla_{\Gamma_h} v_h} dS_h + \int_{\Gamma_h} V_h u_h v_h dS_h.$$

With the following in mind:

$$\int_{\Gamma_h} f dS_h = \sum_{T \in T_h} \int_T f dS$$

That means the assembled matrices are built from real geometric integrals, while the unknown coefficient vector is complex-valued. If V is real, the resulting matrix is Hermitian, which is exactly what you want for the quantum Hamiltonian

So expanding in the nodal basis:

$$\psi_h = \sum_{j=1}^N \Psi_j \varphi_j.$$

the test functions are still the **same finite element basis functions**, but now you allow **complex coefficients**. In practice, for your sphere FEM with standard P_1 elements, the hat functions ϕ_i themselves are real-valued geometric shape functions, and the trial/test space is their complex span.

you should think of

$$V_h^{\mathbb{C}} = \text{span}_{\mathbb{C}}\{\phi_1, \dots, \phi_N\}.$$

That means any discrete function has the form

$$u_h = \sum_{j=1}^N U_j \phi_j, U_j \in \mathbb{C},$$

So if you choose $v_h = \phi_i$, the formula for the matrix entries looks almost identical to the real-valued case:

$$A_{ij} = \frac{\hbar^2}{2m} \int_{\Gamma_h} \nabla_{\Gamma_h} \phi_j \cdot \nabla_{\Gamma_h} \phi_i dS + \int_{\Gamma_h} V_h \phi_j \phi_i dS,$$

provided V_h is real.

That is why the matrix entries coming from geometry are often real, even though the coefficient vector Ψ is complex.

you then get a linear system

$$A\Psi = F,$$

with entries

$$A_{ij} = \frac{\hbar^2}{2m} \int_{\Gamma_h} \langle \nabla_{\Gamma_h} \varphi_j, \nabla_{\Gamma_h} \varphi_i \rangle dS + \int_{\Gamma_h} V_h \varphi_j \bar{\varphi}_i dS,$$

$$F_i = \int_{\Gamma_h} f_h \bar{\varphi}_i dS.$$

Take a single triangle $T \subset \Gamma_h$ with vertices x_a, x_b, x_c .

Now take one specific triangle

$$T = [x_a, x_b, x_c].$$

Any point $x \in T$ can be written in barycentric form

$$x = \lambda_1 x_a + \lambda_2 x_b + \lambda_3 x_c, \quad \lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \lambda_i \geq 0,$$

and on this triangle the three local hat functions are simply

$$\phi_a|_T = \lambda_1, \phi_b|_T = \lambda_2, \phi_c|_T = \lambda_3.$$

So on each triangle, hat functions are just barycentric coordinates. More explicitly

Define edge vectors

$$e_1 = x_b - x_a, e_2 = x_c - x_a,$$

Let $\varphi_a, \varphi_b, \varphi_c$ be the nodal basis functions associated with these vertices, restricted to T .

and define the affine map from the reference triangle

$$\hat{T} = \{(\xi, \eta): \xi \geq 0, \eta \geq 0, \xi + \eta \leq 1\}$$

to the physical triangle by

$$\mathbf{F}(\xi, \eta) = \mathbf{x}_a + \xi \mathbf{e}_1 + \eta \mathbf{e}_2.$$

The Jacobian matrix is

$$J = [\mathbf{e}_1 \ \mathbf{e}_2] \in \mathbb{R}^{3 \times 2},$$

and the induced 2×2 metric on the flat triangle is

$$G = J^T J = \begin{pmatrix} \mathbf{e}_1 \cdot \mathbf{e}_1 & \mathbf{e}_1 \cdot \mathbf{e}_2 \\ \mathbf{e}_2 \cdot \mathbf{e}_1 & \mathbf{e}_2 \cdot \mathbf{e}_2 \end{pmatrix}.$$

On the reference triangle, the three basis functions are

$$\hat{\phi}_1(\xi, \eta) = 1 - \xi - \eta, \quad \hat{\phi}_2(\xi, \eta) = \xi, \quad \hat{\phi}_3(\xi, \eta) = \eta.$$

to grasp what's going on, we're using barycentric coordinates, as coordinates of the flat triangle to approximate our operators on this mesh, in other words look at what happens when we consider the gradients of each basis vector in their reference:

$$\nabla_{\xi, \eta} \phi_1 = (-1, -1), \quad \nabla_{\xi, \eta} \phi_2 = (1, 0), \quad \nabla_{\xi, \eta} \phi_3 = (0, 1).$$

Because the map is affine, the physical gradients on T are also constant.

The physical tangential gradient of a basis function is

$$\nabla_T \phi_i = J G^{-1} \nabla_{\xi, \eta} \hat{\phi}_i.$$

Then the local stiffness matrix is

$$(K_T)_{ij} = \int_T \nabla_T \phi_i \cdot \nabla_T \phi_j \, dS = |T| (\nabla_T \phi_i \cdot \nabla_T \phi_j),$$

If you also have a potential V , the simplest first implementation is to approximate V as constant on each triangle, for example by $V_T = V\left(\frac{x_a + x_b + x_c}{\|x_a + x_b + x_c\|}\right)$, and then take

$$W_T \approx V_T M_T.$$

That gives the local contribution of the Schrödinger potential term

Similarly, write the local mass matrix $M^{(T)}$ with entries

$$M_{pq}^{(T)} = \int_T \varphi_q \bar{\varphi}_p \, dS,$$

and compute it explicitly (it's the standard 3×3 mass matrix on a triangle).